



Characterization and prediction of the airport operational saturation

Álvaro Rodríguez-Sanz*, Fernando Gómez Comendador, Rosa Arnaldo Valdés,
Javier A. Pérez-Castán

Universidad Politécnica de Madrid (UPM), School of Aeronautical and Space Engineering (ETSIAE), SATAA Department, Pza. Cardenal Cisneros N3, 28040 Madrid, Spain



ARTICLE INFO

Keywords:

Aircraft processes
System saturation
Performance indicators
Delay propagation
Bayesian networks

ABSTRACT

This paper develops a functional analysis of the aircraft flow through the airport operational framework, focusing on the airspace-airside integrated system. In this analysis, we use a dynamic spatial boundary associated with the Extended Terminal Maneuvering Area (E-TMA) concept, so inbound and outbound timestamps can be considered. Aircraft operations are characterized by several temporal milestones, which arise from the combination of a Business Process Model (BPM) for the aircraft flow and the Airport Collaborative Decision Making (A-CDM) methodology. This timestamp approach allows us to study the successive hierarchical tasks. The objective is to establish a taxonomy that classifies the system's capacity to “receive and transmit” aircraft streams with adherence to the expected schedule. By considering the accumulated delay across the different processes and its evolution, several indicators are proposed to evaluate the system's level of saturation and its ability to ensure an appropriate aircraft flow in terms of time-efficiency. Finally, the relationships between the factors that influence the aircraft flow are evaluated to create a probabilistic graphical model, using a Bayesian Network (BN) approach. This model predicts outbound delays given the probability of having different values at the causal control variables. The methodology is developed and validated through a case study at Adolfo Suárez Madrid-Barajas Airport (LEMD): a collection of nearly 34,000 turnaround operations (registered at the peak traffic months of 2016) is used to statistically determine the aircraft path characteristics. The contribution of the paper is twofold: it presents a novel methodological approach to evaluate and predict the system's state at the rotation stage and it also provides insights on the interdependencies between factors influencing performance.

1. Problem statement and motivation

Air transport functionality depends on a complex network architecture, where several facilities, processes and agents are interrelated and interact with each other (Goedeking, 2010). In this large-scale and dynamic system, airports represent the interconnection nodes that help aircraft distribution through the network, enable crew and passenger connectivity and ease transport modal changes (Belobaba et al., 2015).

Potential incidents, failures and delays (due to service disruptions, unexpected events or capacity constraints) may propagate throughout the different nodes of the network, making it vulnerable (Beatty et al., 1999). This situation has led to system-wide congestion problems and has worsened due to the strong growth in the number of airport operations during the last decades (Pyrgiotis et al., 2013).

The economic cost of congestion in this interconnected and sometimes overscheduled network of airports and aircraft is enormous: direct costs due to flight delays in Europe reached €1.25 billion during

2010, according to Cook and Tanner (2014). In the United States, during 2007, the directly or indirectly costs originated by delays were around \$32.9 billion (Ball et al., 2010). Furthermore, delays have also a substantial impact on the schedule adherence of airports and airlines, passenger experience, customer satisfaction and system reliability (Bazargan, 2010; Campanelli et al., 2016; Cook et al., 2012b; Deshpande and Arkan, 2012; Jetzki, 2009).

A significant portion of delay generation occurs at airports, where aircraft connectivity acts as a key driver for delay propagation (Gopalakrishnan et al., 2016). During 2015, in the EUROCONTROL Statistical Reference Area, the share of reactionary delay (due to the late arrival of the aircraft on its previous leg) was 45% of total delay minutes (5.1 min of the 11.3 min average delay per flight) and airline related delays (a category that includes crew connections) accounted for another 28% of delay minutes (EUROCONTROL, 2017). Moreover, 33% of all delayed flights in the United States in 2016 were due to reactionary delays (39% of total delay minutes), while airline delay was

* Corresponding author.

E-mail addresses: alvaro.rodriguez.sanz@upm.es (Á. Rodríguez-Sanz), fernando.gcomendador@upm.es (F.G. Comendador), rosamaria.arnaldo@upm.es (R.A. Valdés), javier.perez.castan@upm.es (J.A. Pérez-Castán).

<https://doi.org/10.1016/j.jairtraman.2018.03.002>

Received 13 August 2017; Received in revised form 11 March 2018; Accepted 12 March 2018

Available online 30 March 2018

0969-6997/ © 2018 Elsevier Ltd. All rights reserved.

the cause for another 27% of delayed flights and 33% of total delay minutes (United States Department of Transportation, 2017). “Rotation” (flight cycle through the airport and its surrounding airspace, from inbound to outbound processes) is therefore the stage that has the greatest influence on punctuality within the entire air transport network, and accumulates its impact over the day (Beatty et al., 1999). Hence, this paper focuses on the rotation stage and analyzes the aircraft flow through the airport operational environment, which is the dominant mechanism by which delays propagate through the air transport network (Rebollo and Balakrishnan, 2014).

The evolution of a flight can be described as a sequential flow of events or processes (Katsaros et al., 2013; Schaefer and Noam, 2003; Wu and Caves, 2004a). Each of these events occurs consecutively, and if any of them gets delayed, this may result in subsequent processes also being delayed (unless certain buffers or “slacks” are added into the times allocated to the completion of certain events) (Cook, 2007; Hirst, 2008). In order to analyze the evolution of the aircraft flow and the potential delays in the successive phases, this paper follows a “milestone approach” by assigning completion times to each event. This view, in line with the Airport Collaborative Decision Making (A-CDM) method (EUROCONTROL et al., 2012), allows us to understand the operational performance and the potential saturation of the system. Saturation is here understood as the lack of capacity at the airport-airspace system to “receive and transmit” aircraft flows on time (schedule adherence).

In this analysis we use a dynamic spatial boundary associated with the Extended Terminal Maneuvering Area (E-TMA) concept, which allows us to consider inbound and outbound timestamps. This management boundary (airport centric limit of 200–500 NM) has already been implemented at multiple airports, with a horizon that ranges from around 190 NM for Stockholm to 250 NM for Rome and 350 NM for Heathrow (Bagieu, 2015). The E-TMA (and not just the basic on-ground turnaround path at the airport that connects inbound and outbound flights) is selected in order to integrate delay propagation in the airport system with global delays in the air traffic network. This approach reflects the interaction between airport and airspace processes. The analysis is focused on the aircraft flow-airside operations (Fig. 1).

In time, we restrict actions to a tactical phase (day of operations) in order to consider the primary and initial inefficiencies of the system.

The main objectives of the study are: (a) to analyze the aircraft's flow of processes, in order to define metrics and indicators that enable airport operators and policy makers to assess the system's state (in terms of time-saturation); and (b) to generate a practical probabilistic model that predicts the outbound delay given different explanatory variables.

2. Background and contribution

This paper revises three main topics: the airport-airspace integrated flow of an aircraft, the propagation of delays through the E-TMA processes and the evaluation of the system's efficiency in terms of time-saturation.

A review of the literature about airport-airspace integration illustrates that several prior studies have dealt with the importance of connectivity at airports (Dobruszkes et al., 2017; Fageda and Flores-Fillol, 2016; Redondi and Gudmundsson, 2016; Suau-Sanchez et al., 2016). This paper revises the linkage between inbound and outbound flights by assessing the aircraft operational flow (turnaround integration in the air transport network). This approach is in line with past analyses (Fricke and Schultz, 2009; Katsaros et al., 2013; Oreschko et al., 2014) and with the SESAR's “Airport Transit View” concept (SESAR, 2014a). Our main contribution in this field is the construction of a business process model (BPM) that shapes the airport-airspace integration, by extending the spatial scope to the E-TMA boundaries. Besides, the statistical characterization of the different processes enables us to understand the particularities of the rotation stage.

Regarding delay propagation through the air transport system, a large number of studies showed the complexity of the network (Ciruelos et al., 2015; Cook et al., 2015a; Pyrgiotis et al., 2013) and the potential impact of delays on the system's reliability (Abdelghany et al., 2008, 2004; Delgado and Prats, 2014; Nash, 2008; Prats and Hansen, 2011). Delay propagation is a global process fostered by relationships inside the network: disruptions in one part of the system can disseminate to many others (Gopalakrishnan et al., 2016; Jetzki, 2009). Therefore, network analysis provides a global view of the transmission process (Cook et al., 2015b). Nevertheless, a significant portion of these propagations (45% in 2016 according to EUROCONTROL (2017)) occurs at airports (i.e. the nodes of the system): “rotation” (delayed flight cycles)

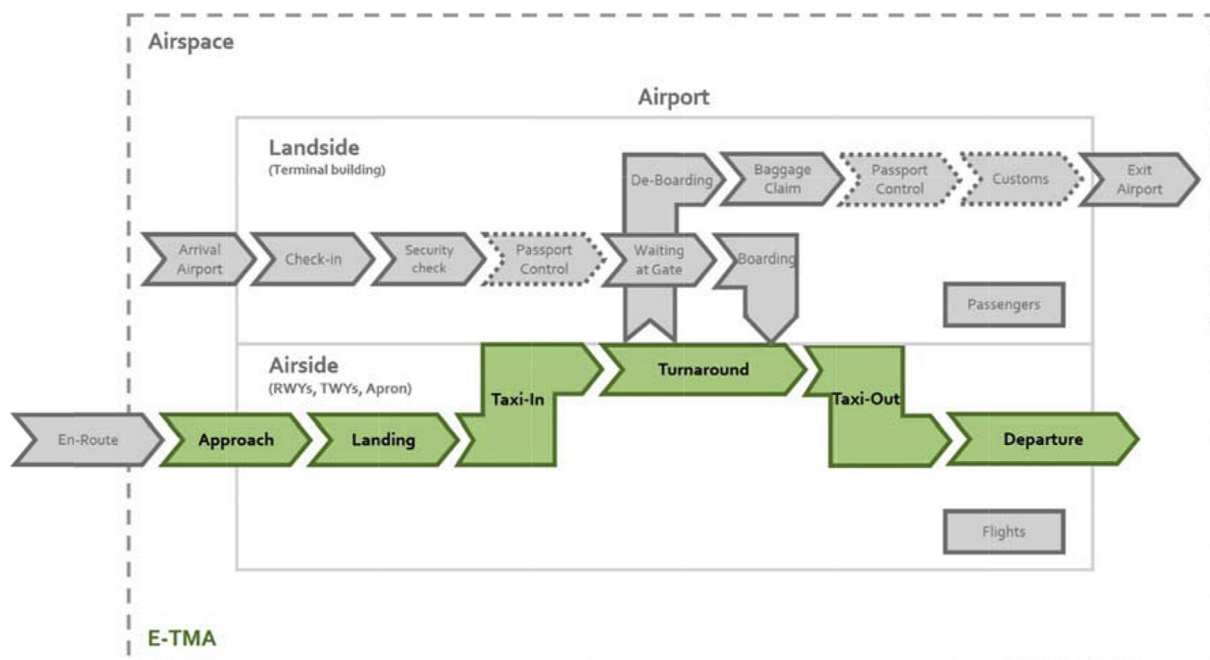


Fig. 1. Spatial scope of the problem.

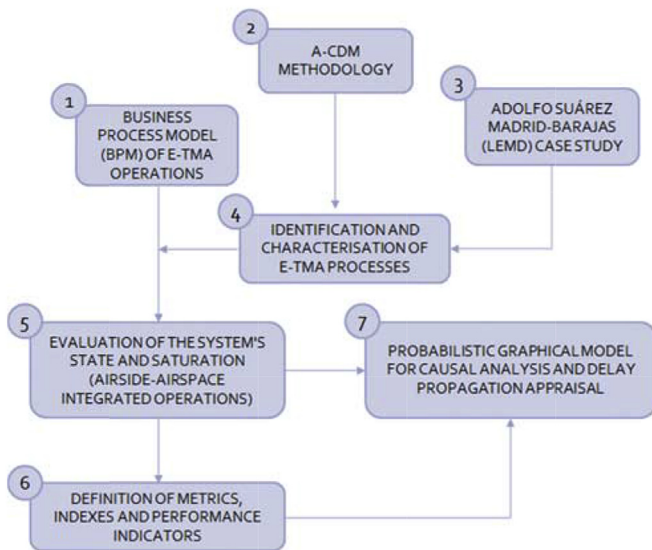


Fig. 2. High level methodology for the analysis.

is the stage that has the greatest impact on punctuality within the entire air transport network. However, delay propagation affecting internal E-TMA and airport processes has received less attention than the effect on the whole system (Fricke and Schultz, 2009; Janić, 1997; Marintseva et al., 2015; Norin et al., 2012; Wu, 2008; Wu and Caves, 2004b). The contribution of our study relies on adjusting the spatial scope of the problem to the rotation stage (integrating airside and airspace operations) and analyzing the potential consequences of its congestion. Neyshabouri and Sherry (2014) concluded that, once different initiatives have successfully decreased airborne delay over the two last decades, the next opportunity for time-efficiency improvement rely on airport surface operations. We extend this objective to the E-TMA framework, in order to cohesively integrate airport (airside) and air traffic operations.

The inherent complexity of the delay propagation problem and the intrinsic challenges in predicting system behavior explain the use of different modelling techniques: queuing theory (Pyrgiotis et al., 2013; Wang et al., 2003), stochastic delay distributions (Tu et al., 2008), propagation trees (AhmadBeygi et al., 2008; Campanelli et al., 2014; Fleurquin et al., 2014), periodic patterns (Abdel-Aty et al., 2007), chain effect analysis (Wong and Tsai, 2012), random forest algorithms (Rebollo and Balakrishnan, 2014), statistical approaches (Churchill et al., 2007), non-linear physics (Deutschmann, 2012), phase changes (Belkoura and Zanin, 2016) and dynamic analysis (Gopalakrishnan et al., 2016). In this paper, delay propagation patterns and influence variables are characterized using a Bayesian Network (BN) approach, including stochastic parameters to reflect the inherent uncertainty of the aircraft flow performance at the E-TMA. Several studies (Buldyrev et al., 2010; Liu and Ma, 2008; Zio and Sansavini, 2011) demonstrate the utility of graph theory and BNs as a methodology for modelling the diffusion of events and incidents from a node-level to a system-level (interdependence of multiple factors). Moreover, Liu et al. (2008), Liu and Wu (2013) and Xu et al. (2005) confirmed that BNs can explain how subsystem-level causes propagate to provoke system-level effects, specifically focusing on how delays at an origin airport propagate to create delays at a destination airport.

Regarding time performance metrics and efficiency indexes, ICAO (2009) established the foundations with regard to punctuality and predictability indicators in aviation. The generally accepted key performance indicator (KPI) for operational air transport performance is “punctuality”, which can be defined as the proportion of flights delayed by more than 15 min compared to the published schedule (Jetzki, 2009). The 15-min threshold for defining arrival and departure delay

has historically been common to both Europe and the US (Cook et al., 2012a; Sherry et al., 2013). SESAR's Performance Targets (SESAR, 2014b) significantly refined this approach to delay measurement, by developing new parameters, indicators and targets. Cook et al. (2015b, 2012a) showed that, although delay propagation remains a significant and costly operational challenge to ATM (Air Traffic Management), there is a substantial absence of metrics that specifically measure this problem. Findings settled by Cook et al. (2015a), Guldung et al. (2013) and Simić Krstić and Babić (2015) developed a framework for complexity and new metrics related to airports and ATM.

Moreover, Trapote-Barreira et al. (2016) analyzed the applicability of a series of network connectivity and concentration indexes, in order to typify complex airline network configurations. Other studies have proposed performance indicators for airport operations (Kosanke and Schultz, 2015), delay model calibration (Churchill et al., 2008), delay propagation networks (Belkoura and Zanin, 2016), aircraft delay and capacity thresholds (Rakas, 2014), and airline schedule adherence and reliability (Wu, 2005). For the purposes of this paper, different metrics are formulated to evaluate the system's state and its ability to ensure an appropriate aircraft flow in terms of time-saturation. Our main contribution in this topic is a taxonomy that classifies the system's capacity to “receive and transmit” the expected aircraft flows with time efficiency.

3. Methodology and discussion

The analysis is divided into two steps:

Firstly, we develop a theoretical appraisal of the aircraft operation within the E-TMA, by characterizing the processes and structuring the different timestamps. We generate a business process model, which is combined and quantified with the A-CDM milestone approach. This provides us with a conceptual framework for the practical analysis of the rotation flow.

The second part of the analysis is developed with a practical case study at Adolfo Suárez Madrid-Barajas (LEMD) Airport. We assess the system's state and saturation by appraising its time efficiency performance. This is achieved by statistically evaluating the processes that were previously recorded in the first section and also by defining metrics, indexes and indicators that represent the delay behavior. After that, a probabilistic model is assembled considering the interactions between the different delay explanatory variables. This model offers information about the system's state by predicting outbound delays. The method is applied to the data of the case study in order to validate its usefulness.

Fig. 2 shows the logic behind the analysis.

3.1. Model for airside-airspace integrated operations (rotation stage)

The aircraft rotation stage is usually the critical node for the air transport network: incoming aircraft continue on the subsequent legs of their planned itineraries and crew members and passengers may connect to other flights or other transport modes (Bazargan, 2010; Gopalakrishnan et al., 2016; Wu, 2012).

The aim of the study is the description of the “visit” of an aircraft to the E-TMA, as an extension of the SESAR's “Airport Transit View” concept (Fig. 3). This “visit” consists basically of three separate sections (SESAR, 2014a):

- The final approach and inbound ground section of the inbound flight.
- The turnaround process section in which the inbound and the outbound flights are linked.
- The outbound ground section and the initial climb segment of the outbound flight.

Developing the conceptual structure of the aircraft flow within the

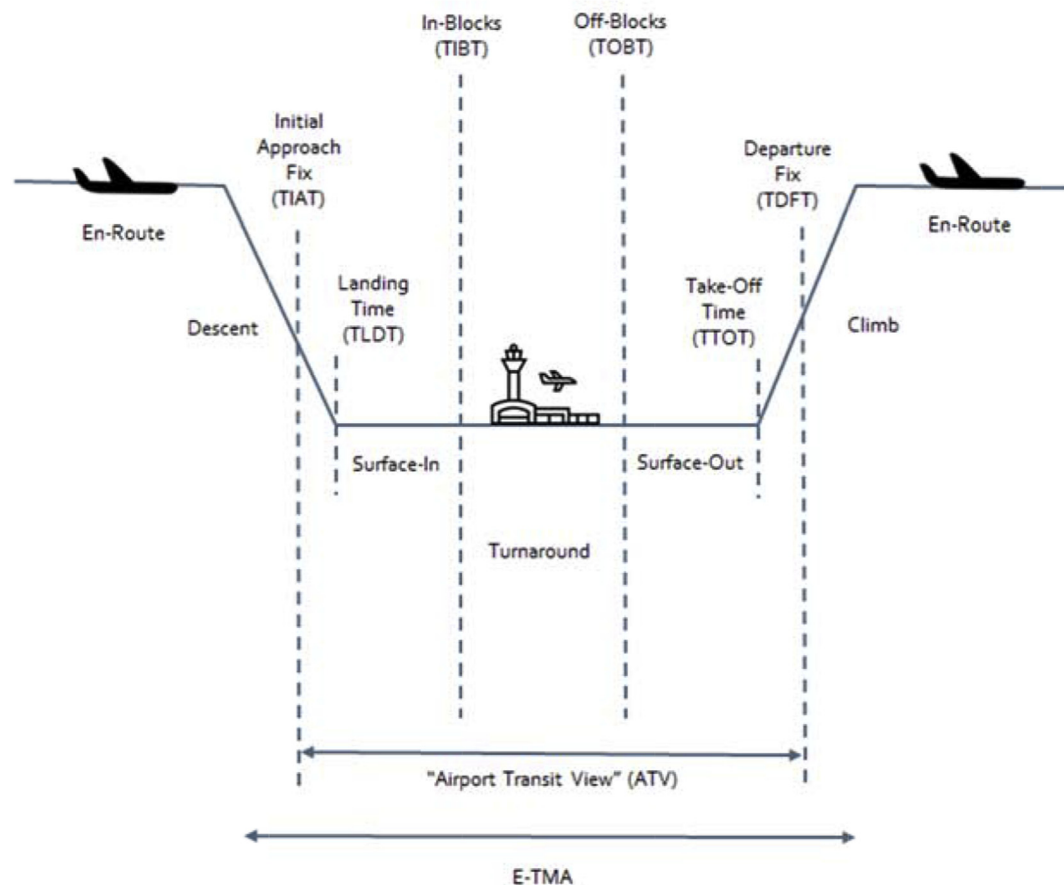


Fig. 3. Extension of the Airport Transit View concept. Source: adapted from (SESAR, 2014a).

E-TMA requires input from various sources and consists of four main steps (Wilke et al., 2014):

- 1 The first step is a review of relevant literature and existing aircraft flow models (Ashford et al., 2013; Bazargan, 2010; Kazda and Caves, 2007; Schmitt and Gollnick, 2016; SESAR, 2014a; Wu, 2012).
- 2 Next, a hierarchical task analysis is developed (Kirwan and Ainsworth, 1992). This appraisal follows a top-down approach that incorporates several sources of information in order to give a detailed understanding of the processes:
 - a) Analysis of operations manuals (IBERIA, 2015, 2014), standards and procedures (IATA, 2015; ICAO, 2016a, 2006a, 2006b, 2006c).
 - b) Observations at Adolfo Suárez Madrid-Barajas Airport (LEMD) during 2016.
 - c) Structured communications with relevant stakeholders (Table 1).
3. The previous steps lead to an initial process model.
4. Finally, the initial model is refined and validated with the help of

subject-matter experts (Table 1).

Fig. 4 depicts the methodology (Wilke et al., 2014) to create a Business Process Model (BPM) of the aircraft flow at the E-TMA (rotation stage).

We employ Unified Modelling Language (UML) to graphically represent the BPM. UML is a visual modelling language that can be used to create a pattern of a system (Engels et al., 2005). The designed conceptual structure for the airport-airspace integrated operations is basically a UML sequence diagram (Fig. 5). This model is then confronted to the operational milestones defined by the A-CDM methodology. A-CDM aims at improving the overall efficiency of airport operations by optimizing the use of resources and enhancing the predictability of events. It focuses especially on aircraft turnaround and pre-departure sequencing processes (EUROCONTROL et al., 2012). The milestones approach main goal is to achieve common situational awareness by tracking the progress of a flight from the initial planning to the take off. It defines and describes “timestamps” to enable close monitoring of significant events (EUROCONTROL et al., 2012). The A-CDM implementation is proven to provide several benefits in areas such as delay/capacity management, resource allocation, operational predictability, noise abatement and fuel savings for all major stakeholders, including airlines, ground handlers and the network manager (EUROCONTROL, 2016).

Fig. 6 and Table 2 show the set of selected milestones along the progress of the flight at the A-CDM concept. See Annex I for the meaning of the acronyms.

By combining the BPM and the milestone approach, Fig. 7 shows a conceptual diagram for the E-TMA operational environment (airport-airspace stage). This diagram allows us to:

Table 1
List of informants, interviewees and contributors.

Organization	Stakeholder
AENA - Spanish Airport Authority and Airport Manager	Airport operator
IBERIA - Member of International Airlines Group (IAG)	Airline
ENAIRES - Spanish Air Navigation Service Provider	Air Navigation Services Provider (ANSP)
IBERIA Airport Services	Ground Handling Agent
DGAC - Spanish General Directorate of Civil Aviation (a public body answerable to the Ministry of Public Works)	Policy maker - Regulator

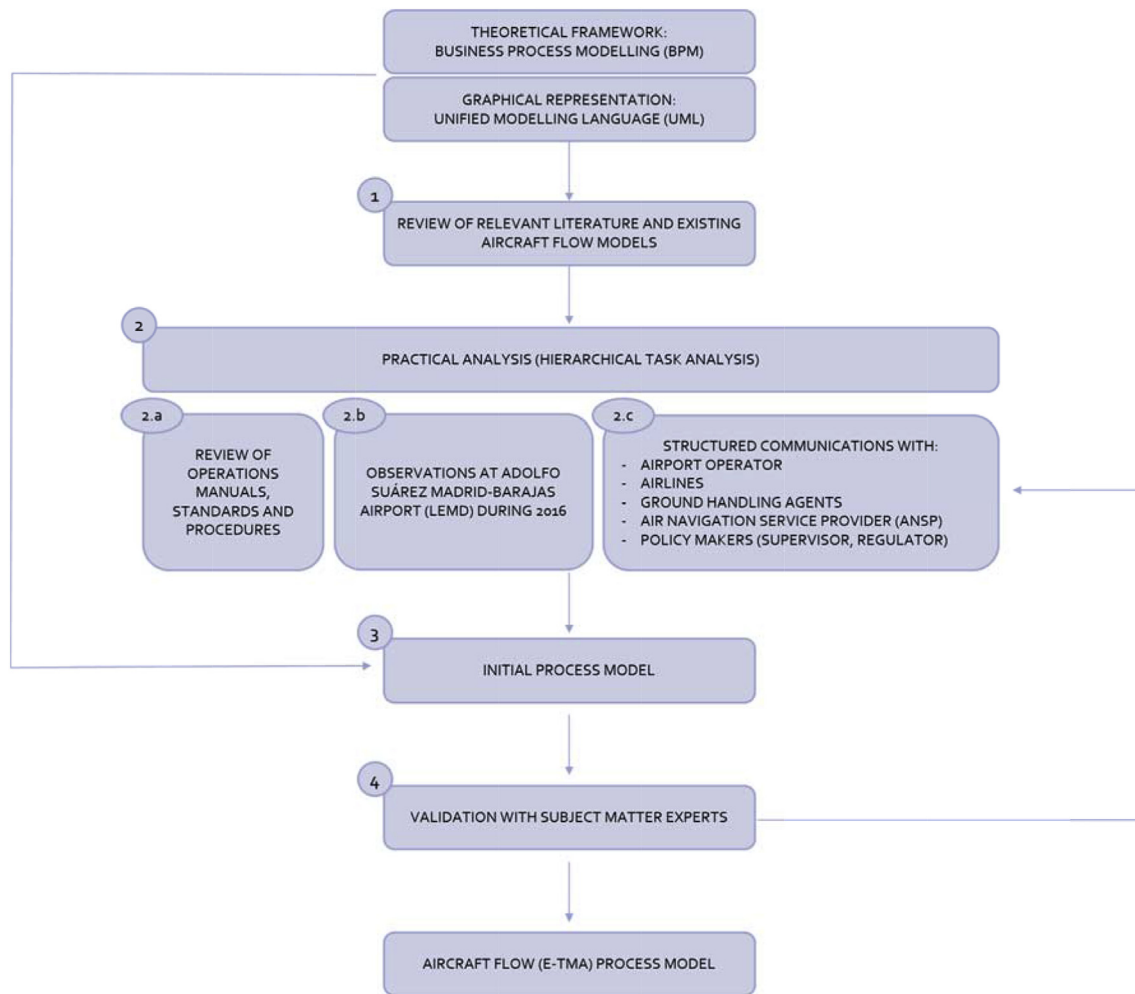


Fig. 4. Methodology to create the BPM of the aircraft flow. Source: adapted from (Wilke et al., 2014).

- Determine significant events in order to track the progress of the flight (arrival, landing, taxi-in, turnaround, taxi-out and departure) and the distribution of these key events as milestones.
- Ensure linkage between arriving and departing flights.
- Assess time efficiency performance, which is measured for each milestone or between two milestones.
- Enable early decision making when there are disruptions to an event.

3.2. Characterization of the system's level of saturation

The analysis of the system's level of saturation is developed with a case study at Adolfo Suárez Madrid-Barajas Airport (LEMD). The methodology could be nevertheless applied to other airports by adjusting the model to the infrastructure characteristics, the operational pattern and the available data.

Fig. 8 shows the structure of LEMD, with four runways (36L-18R, 36R-18L, 32L-14R, 32R-14L), four terminal areas (T123, T4T4S, general aviation and cargo) and 163 aircraft parking stands (ENAI, 2018).

LEMD is a large airport in terms of passengers and operations: it handled 50,420,583 passengers and 378,150 aircraft movements (including arrivals and departures) in 2016, according to the airport manager and operator (AENA, 2017). Therefore, there were sufficient operations (and a continuous demand) during the observation period: July and August, which were the peak months in terms of traffic (4,906,164 passengers and 34,476 aircraft movements for July and

4,793,527 passengers and 33,202 aircraft movements for August, according to AENA (2017)).

The operational preferential configuration at LEMD is called north configuration, with arrivals from runways 32L/32R and departures from runways 36L/36R. The non-preferential configuration (south) presents arrivals from runways 18L/18R and departures from runways 14L/14R. Night flights (between 23:00 and 07:00 local time) use 32R (arrivals) – 36L (departures) for north configuration and 18L (arrivals) – 14L (departures) for south configuration (ENAI, 2018).

LEMD can be considered a major connecting hub, where several airlines have a significant presence at the airport with a business model for transferring passengers between arriving and departing flights (Rakas, 2014).

Regarding registered delays at LEMD, as a departure airport it ranked number 34 among the top delay affected departure airports in Europe during 2016, with 11.4 min of average delay per departure (an increase of 13% since 2015), 26.5 min of average delay per delayed departure and 43% of delayed departures (EUROCONTROL, 2017). As an arrival airport, LEMD ranked number 19 among the top delay affected arrival airports in Europe during 2016, with 11.5 min of average delay per flight (an increase of 20% since 2015), 28.2 min of average delay per delayed arrival and 40.9% of delayed arrivals (EUROCONTROL, 2017). These values make LEMD a suitable airport for characterizing and analyzing delay and saturation patterns.

The observation period corresponds to July and August 2016, when 67,678 aircraft movements (arrivals and departures) were registered at LEMD. Therefore, a collection of nearly 34,000 turnaround operations

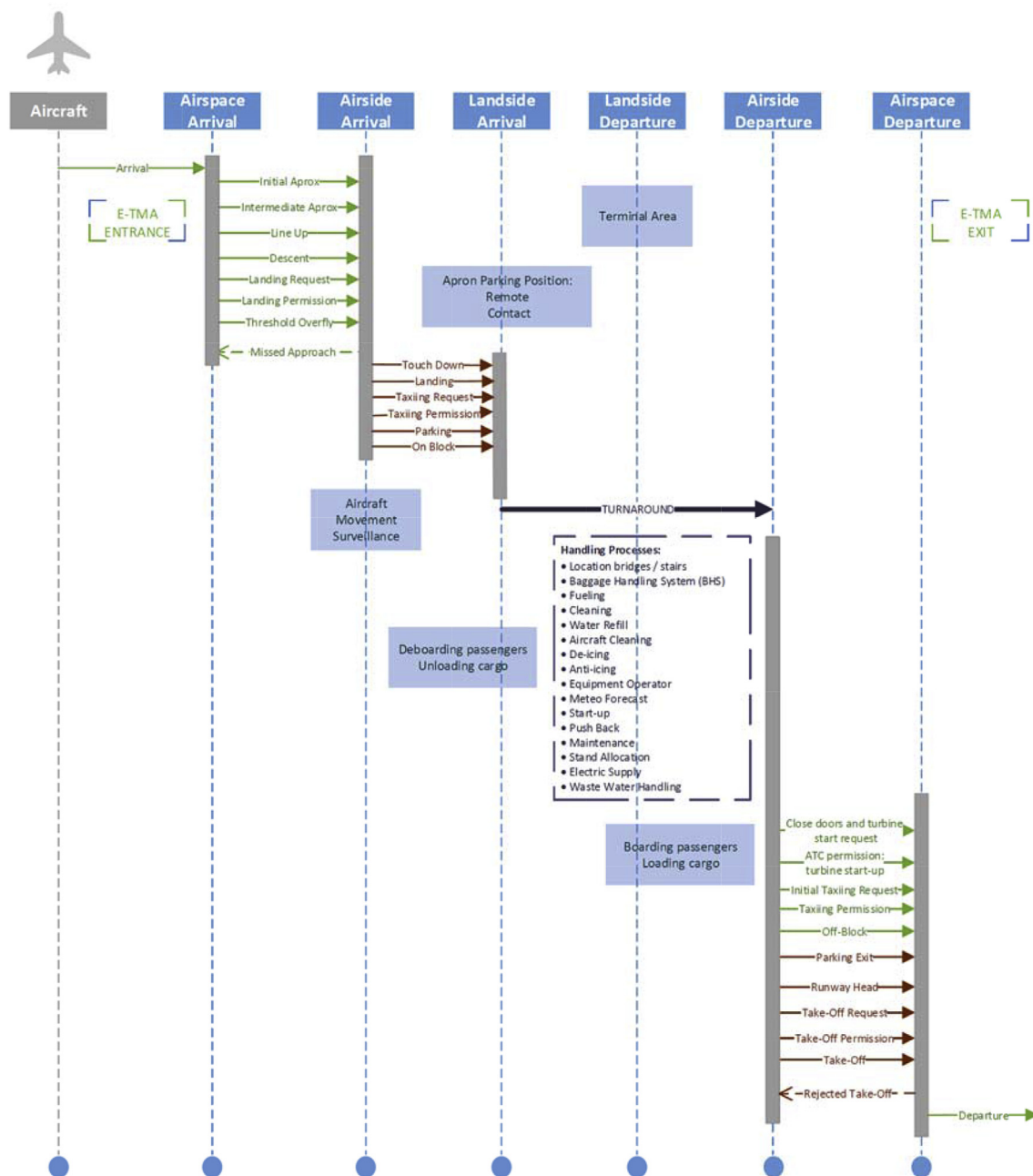


Fig. 5. UML for the BPM of the aircraft flow (airport-airspace integrated operations).

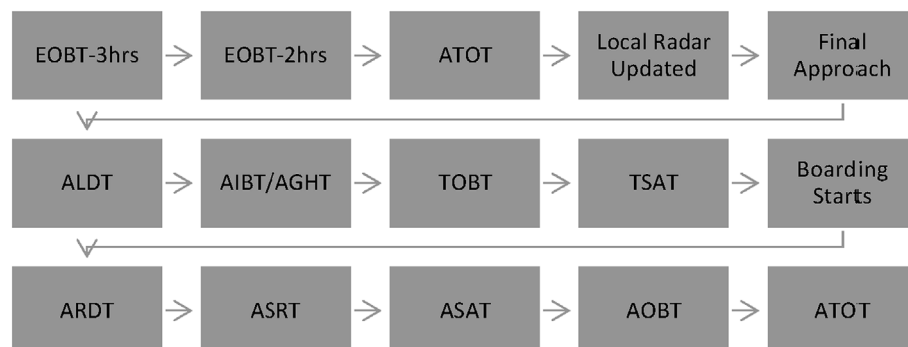


Fig. 6. Milestones reflecting the progress of the flight (A-CDM concept). Source: adapted from (EUROCONTROL et al., 2012).

Table 2
Selected milestones along the process of the flight (A-CDM concept). Source: adapted from (EUROCONTROL et al., 2012).

Milestone number	Milestone meaning
M1	EOBT (Estimated Off-Block Time)-3 h
M2	EOBT (Estimated Off-Block Time)-2 h
M3	Take Off from Outstation
M4	Local Radar Update
M5	Final Approach
M6	ALDT (Actual Landing Time)
M7	AIBT (Actual In-Block Time)
M8	Actual Ground Handling Starts
M9	TOBT (Target Off-Block Time)
M10	TSAT (Target Start-Up Approval Time)
M11	Boarding Start
M12	ARDT (Aircraft Ready Time)
M13	ASRT (Actual Start-Up Request Time)
M14	ASAT (Actual Start-Up Approval Time)
M15	AOBT (Actual Off-Block Time)
M16	ATOT (Actual Take Off Time)

is used to describe the aircraft flow characteristics, through a statistical analysis of the processes. The size of the sample allows us not only to perform representative post operational analysis but also to develop reliable predictions regarding schedule adherence and to study the interdependencies between the turnaround processes and the different involved variables.

The data preparation phase covers all activities required to arrange the final dataset from the initial raw meteorological, radar and airport operational data, including locating and refining erroneous measurements. As shown in Table 3, the final dataset includes information about airport infrastructure allocated to each operation, airline and aircraft characteristics, flight details, route origin and destination, operational times (milestones and processes' lengths), regulations, delay causes and meteorological features.

Data regarding the aircraft and the flight (type, call sign and registration number) enable us to link the inbound and outbound movements, assessing their “turnaround” operation (i.e. trace the airport-airspace integrated operations).

Fig. 9 shows the operational time milestones that are included in the dataset (consistent to the previously developed functional model, which was a combination between the BPM and the A-CDM approach). The timestamps are represented chronologically (not to scale). See Annex I for the meaning of the acronyms. Fig. 9 also illustrates the five main delay measures that are considered in the analysis of the system's time efficiency performance and level of saturation (their definition can be seen in Table 5):

- Arrival delay (reactionary delay)
- Taxi-in delay
- Turnaround delay
- Taxi-out delay
- Departure delay

Departure delay is appraised as the sum of arrival upstream delay (reactionary) and the aggregated delay at the rotation stage: system delay (primary delay), which is composed of taxi-in, turnaround and taxi-out delay. Therefore, four mutually exclusive and complementary stages are evaluated in order to characterize the system's saturation: arrival, taxi-in, turnaround and taxi-out.

Consequently, departure delays result from various reasons, such as “inherited” arrival lateness, delayed ground processes and/or disturbed ground operations. Interdependencies exist and may affect the delay chain: existing reactionary delay may result in an even increased follow-up delay due to scarce resources at the airport (Wu, 2012).

We also consider four more measures that are already included in the main delays: (a) excess in approach queuing time (included in arrival delay); and (b) start-up, (c) push and (d) ATFCM delay (included in turnaround delay). ATFCM delay is due to flow and capacity

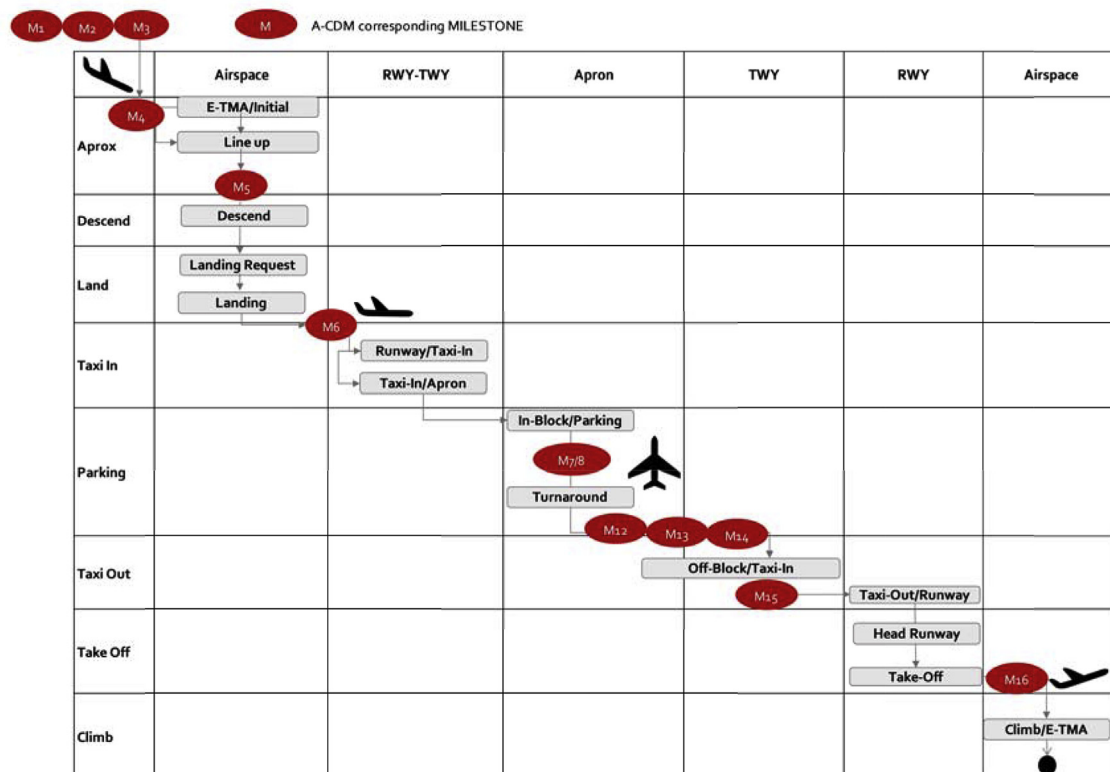


Fig. 7. Combination of the BMP for the airport-airspace integrated operations and the A-CDM concept.

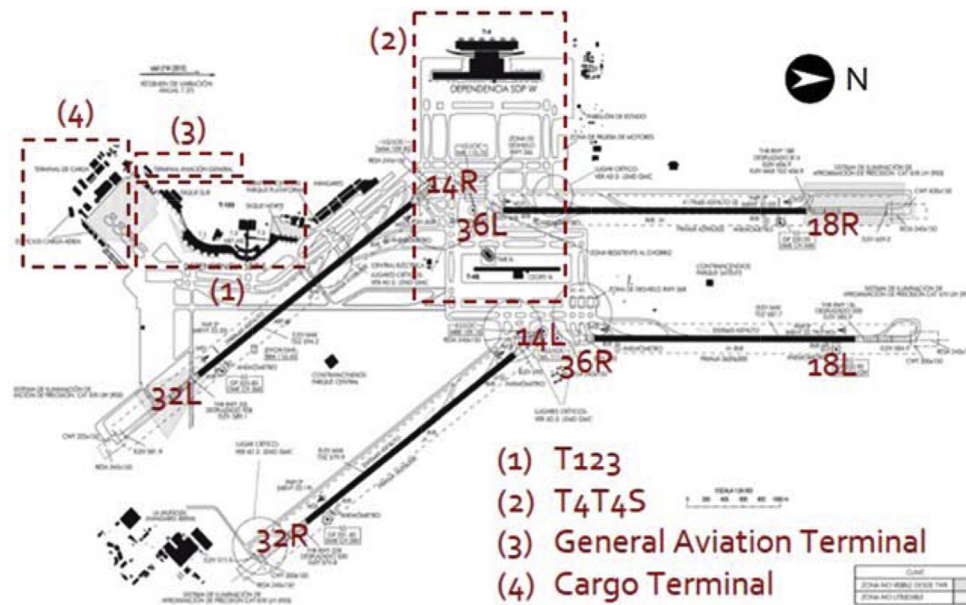


Fig. 8. Madrid Airport (LEMD) functional structure (ENAI, 2018).

regulations: aircraft are hold on ground, preventing them to encounter airborne delays during which fuel is burnt and emissions are produced (Sandrine et al., 2007).

For the purpose of our study, delays are here defined as “schedule delays”: the difference between a planned time and the actual time. Despite the use of “delay”, “schedule delay” can refer to a difference in either the early or late direction (Small, 1982). Therefore, schedule delays can be positive or negative.

Departure delay is defined as the sum of arrival delay + taxi in/out delay + turnaround delay. If the turnaround process is efficient enough (or if its allocated time is sufficiently large to ensure that the operation ends ahead of schedule), turnaround delay can be negative (note that schedule delays can refer to a difference in either the early or late direction and therefore can be positive or negative). Hence, even with “positive” arrival delay and taxi in/out delays, departure delay could be zero (or even negative) if turnaround delay is “negative” and

sufficiently large to absorb delays at other stages. Table 4 shows several operational examples from the dataset where the turnaround process was able to absorb arrival and taxi delays.

It should be noted that there is a difference between the aircraft scheduled turnaround time and the actual turnaround time that goes beyond delay. It is related to the concept of schedule contingencies or “buffers”. There are three main types of buffers (Cook, 2007):

- “Off-block” buffer, which is added to the time allocated to get from gate to gate in order to absorb any off-block delays, such as runway sequencing and airborne delay (e.g. arrival management).
- “At-gate” buffer, defined as the additional time built into the schedule specifically to absorb delay whilst the aircraft is on the ground and to allow recovery between the rotations of aircraft (although it may be necessary to wait for connecting passengers or for a crew change).

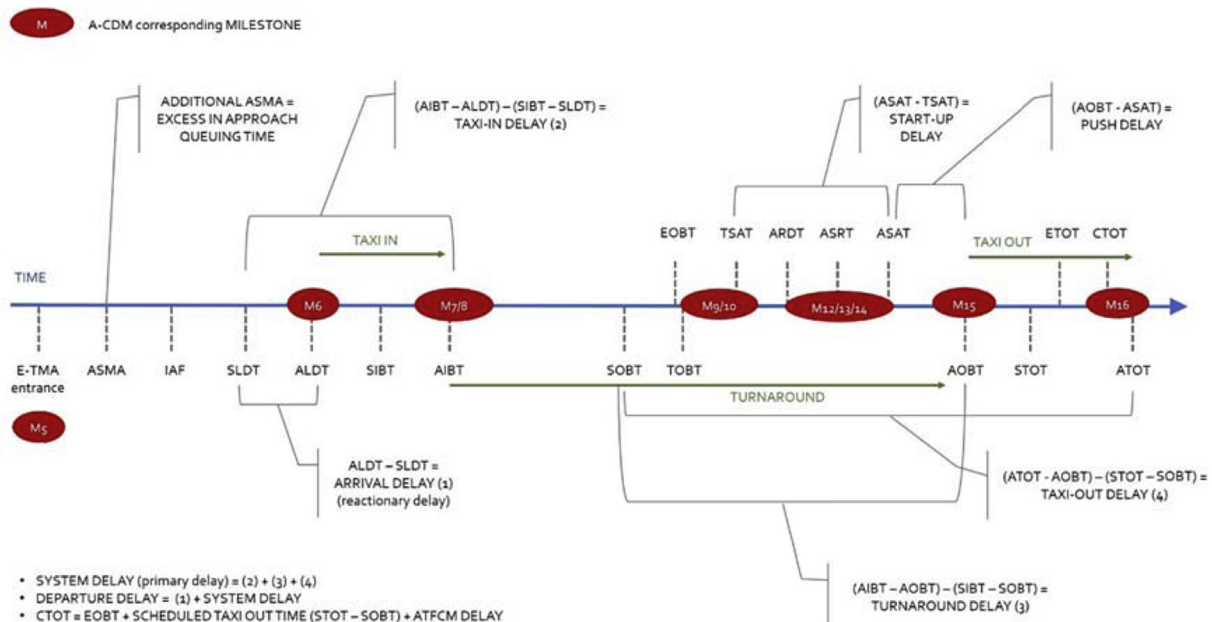


Fig. 9. Milestones and delays considered. Source: adapted from (EUROCONTROL et al., 2012).

Table 3
Information included in the final dataset.

Type of data	Information included in the dataset (obtained from raw data or derived from it)
Airport infrastructure	Runway and stand use (terminal area). Runway declared capacity (arrivals, departures and total)
Airline	Operator, type (Low Cost Carrier/Network/Cargo/General Aviation) and associated handling agent
Aircraft	Model, wake turbulence category (High/Medium/Low), size (Narrow/Wide body) and registration number
Flight	Flight number, type (commercial or private) and Air Traffic Control (ATC) call sign
Route	Origin and destination, category (Domestic/European/Long Haul)
Operational times (see Fig. 9) & regulations	Date, aircraft processes' milestones (from the E-TMA entrance to its exit: approach, on-ground turnaround and climb, see Fig. 9), timestamps (schedule adherence), processes' lengths, holding patterns, aircraft separation, number of aircraft queuing for the inbound traffic flow and ATFCM (Air Traffic Flow and Capacity Management) regulations
Throughput (airspace and airspace)	E-TMA throughput (movements per hour), runway throughput (movements per hour)
Delay causes	Delay causes according to the codes developed by IATA (2015)
Meteorology	Wind (direction and intensity), visibility, RVR (runway visual range), clouds (type and amount), temperature, atmospheric pressure and presence of fog

- “Slot” buffer or slack time, due to the availability of airport slots: waits imposed upon the airline by factors that are essentially exogenous to its scheduling.

Therefore, if we consider the turnaround time as the *actual time required* to perform the different on-ground processes (deplane inbound passengers and their bags, service and refuel the aircraft, board the outbound passengers and their bags), the difference between actual and scheduled timestamps [(AIBT-AOBT) - (SIBT-SOBT)] should be called “excess in turnaround time” rather than “turnaround delay” (in order to consider the existence of buffers).

We analyze the operational pattern of the airport through a statistical characterization of the different processes that were previously identified with the BPM and the A-CDM milestone approach. The aim is to assess the system's level of saturation and its ability to “receive and transmit” aircraft flows with adherence to the expected schedule. Time efficiency performance is measured for each milestone (when scheduled and actual timestamps are available) or between two milestones (to assess the length of the process).

Table 5 illustrates the metrics (processes' duration, disturbances at their start/finish time and delays) that are evaluated (given the available data for the case study). It contains the definition of the measure, its importance and influence on the analysis and the basic statistical data. Fig. 10 shows more detailed statistical information of the appraised items, which allows us to characterize and model the different stages (by fitting the duration of the process and its starting time accuracy to a statistical distribution).

Regarding punctuality, each stakeholder will focus on different indicators (Ashford et al., 2013; Cook et al., 2015b; Wu, 2012): (a) air navigation service providers will revise ATFCM delay; (b) airport operators will analyze off-block delay (AOBT - SOBT) and “ATOT - STOT” (with STOT = SOBT + Scheduled taxi-out) as measures for slot adherence and service agreements; (c) passengers will perceive “ATOT - SOBT” as a standard for time-reliability; and (d) airlines will review scheduling adherence to their operational plans.

When analyzing flight delays, it should be noted that causes of short delays are often quite different from causes of long delays

(EUROCONTROL, 2017; Wu, 2005). This is why Fig. 10 pictures “sample outliers”. If Q_1 and Q_3 are the lower and upper quartiles respectively, then we define an outlier to be any observation outside the range: $[Q_1 - k(Q_3 - Q_1), Q_3 + k(Q_3 - Q_1)]$, with $k = 1.5$ (Tukey, 1977). During the analysis, these outliers (that could be due to faulty data or potential non representative operations) might be excluded from the main sample, because of the possibility of biased results. Nevertheless, for items like #13 (start-up delay) which are highly centered at zero with a wide range of variation, these outliers are important for the analysis, as they provide significant operational information.

Each of the elements shown in Table 5 (stages, starting times and delays) is fitted to a probability density function. Six different statistical distributions were found to be candidates: Weibull, Gamma, Beta, Gumbel, F and Normal. For each item's data distribution, all candidates are evaluated for best fit. This is achieved with a Pearson's chi-square test at 95% significance level (confidence interval) and with 7 degrees of freedom (Plackett, 1983). Fricke and Schultz (2009) and Wu (2012) already found this procedure to be efficient when analyzing on-ground processes. When a parametric distribution cannot properly describe the data, we use a Kernel density estimation approach to obtain a non-parametric representation of the probability density function of the variable (Scott, 2015). See Fig. 11 for some examples.

Distribution fitting has four main objectives: (a) to characterize each process' duration and its starting time; (b) to appraise delays and its statistical behavior; (c) to use probability distributions as a tool for dealing with uncertainty in order to make informed decisions and predictions (e.g. cumulative density functions can be used for allocating turnaround and buffer times); and (d) to use probability distributions as inputs for the causal model that is developed later in this paper. As an example, Fig. 12 shows the analysis for the turnaround stage (actual process length and in-block/off-block adherence), as it is found to be fundamental when assessing the system's ability to absorb delays (Cook, 2007).

As already mentioned, fitting turnaround characteristic times to a statistical distribution may help in the definition of an operational buffer or “slack” (balancing trade-offs between schedule punctuality and aircraft utilization), with the final objective of absorbing arrival

Table 4
Examples of delay absorption by the turnaround process.

Date	Airline type	ALDT	ATOT	Arrival delay (min)	Taxi-in delay (min)	Turnaround delay (min)	Taxi-out delay (min)	Departure delay (min)
04/07/2016	LCC	8:19:00	10:40:18	8	2	-23	3	-10
11/07/2016	Network	16:53:00	18:09:02	23	1	-28	0	-4
14/07/2016	Network	12:33:00	15:29:28	69	5	-74	0	0
22/07/2016	Network	13:16:00	15:11:39	26	0	-29	3	0
07/08/2016	LCC	19:20:00	22:18:01	16	0	-21	4	-1
19/08/2016	Network	8:30:00	11:30:00	24	0	-28	0	-4
27/08/2016	LCC	21:19:00	22:06:26	23	0	-27	0	-4
28/08/2016	Network	19:33:00	21:17:12	38	3	-46	0	-5

Table 5
Items appraised regarding the aircraft flow through the airport-airspace integrated operations.

#	Metric	Definition - Importance	Statistical data: mean value (μ) and standard deviation (σ)
1	ALDT-(ETMA entrance time). <i>E-TMA transit time</i>	The actual time spent by a flight between its entry in the E-TMA and the ALDT is called E-TMA transit time.	$\mu = 29.6$ min $\sigma = 10.0$ min
2	ALDT-IAF. <i>Approach time</i>	The actual time spent by a flight between its pass through the Initial Approach Fix (IAF) point (where the initial approach segment of an instrument approach begins) and the ALDT is called approach time.	$\mu = 6.9$ min $\sigma = 1.4$ min
3	Additional ASMA (Arrival Sequencing and Metering Area). <i>Excess in approach queuing time</i> .	The additional ASMA time is a proxy for the average arrival runway queuing time on the inbound traffic flow, during congestion periods at airports. It is the difference between the actual ASMA time of a flight and a statistically determined unimpeded ASMA time based on ASMA times in periods of low traffic demand.	$\mu = 1.3$ min $\sigma = 2.2$ min
4	ALDT-(ASMA entry time at 40 NM upstream). <i>ASMA transit time</i>	ASMA is defined as a virtual cylinder with a 40 NM radius ^(*) around the airport. The actual time spent by a flight between its last entry in the cylinder and the ALDT is called ASMA transit time.	$\mu = 13.6$ min $\sigma = 3.7$ min
5	ALDT-SLDT. <i>Arrival delay</i> .	It represents the upstream arrival delay (reactionary delay), by assessing whether the landing time (timestamp) is developed as scheduled, with lateness or in advance.	$\mu = 8.3$ min $\sigma = 30.5$ min
6	AIBT-ALDT	It represents the taxi-in process length.	$\mu = 8.9$ min $\sigma = 3.6$ min
7	Actual taxi-in duration (AIBT-ALDT) against Scheduled taxi-in duration (SIBT-SLDT). <i>Taxi-in delay</i> .	It represents a measure for evaluating whether the taxi-in operation at the airport is developed as scheduled, with lateness or better than expected (absorbing delay).	$\mu = 0.3$ min $\sigma = 2.2$ min
8	AIBT - SIBT. <i>In-block adherence</i>	It represents a measure for evaluating whether the in-block operation (timestamp) is developed as scheduled, with lateness or better than expected (absorbing delay).	$\mu = 8.6$ min $\sigma = 30.6$ min
9	AIBT-AOBT	It represents the turnaround process duration.	$\mu = 165.6$ min $\sigma = 179.7$ min
10	Actual turnaround time (AIBT-AOBT) against Scheduled turnaround time (SIBT-SOBT). <i>Turnaround delay</i> .	It represents a measure for evaluating whether the turnaround operation at the airport is developed as scheduled, with lateness or better than expected (absorbing delay).	$\mu = 4.8$ min $\sigma = 28.2$ min
11	ASAT-ASRT	It allows assessing the difference in time between the aircraft operator request for start-up and the actual start-up approval permission by the Air Traffic Controller (ATC).	$\mu = 2.4$ min $\sigma = 21.3$ min
12	AOBT-ASAT. <i>Push delay</i> .	It allows assessing the difference in time between the actual start-up approval permission by the Air Traffic Controller (ATC) and the Off-Block operation.	$\mu = 3.2$ min $\sigma = 8.9$ min
13	ASAT-TSAT. <i>Start-up delay</i> .	It allows to understand whether or not there is some delay between the target time for start-up and the actual one.	$\mu = -1.2$ min $\sigma = 16.5$ min
14	AOBT-SOBT. <i>Off-block adherence</i>	It represents a measure for evaluating whether the off-block operation (timestamp) is developed as scheduled, with lateness or better than expected (absorbing delay).	$\mu = 13.4$ min $\sigma = 25.7$ min
15	AOBT-ATOT	It represents the taxi-out process duration.	$\mu = 16.6$ min $\sigma = 5.3$ min
16	Actual taxi-out duration (AOBT-ATOT) against Scheduled taxi-out duration (SOBT-STOT). <i>Taxi-out delay</i> .	It represents a measure for evaluating whether the taxi-out operation at the airport is developed as scheduled, with lateness or better than expected (absorbing delay).	$\mu = 0.9$ min $\sigma = 4.4$ min
17	ATFCM delay	CTOT = EOBT + Scheduled taxi-out duration (SOBT-STOT) + ATFCM delay. A tool used in the European airspace to regulate the flow of air traffic through individual en-route sectors or airports.	$\mu = 3.9$ min $\sigma = 16.9$ min
18	System delay (primary delay)	System delay = Taxi-in delay + Turnaround delay + Taxi-out delay. It allows assessing the relationship between arrival delay and departure delay for the different operations, i.e. the ability of the airport to absorb or mitigate the arrival delay.	$\mu = 5.8$ min $\sigma = 28.6$ min
19	Departure delay (total delay)	Departure delay = Arrival delay + System delay.	$\mu = 14.1$ min $\sigma = 26.5$ min
20	Local delay	Local delay = System delay - ATFCM delay It allows assessing the “aerodrome true delay”, since it does not consider delay due to en-route regulations.	$\mu = 2.0$ min $\sigma = 30.6$ min

(*) Due to the particular operational characteristics of the airspace around LEMD, that shape the distance for holding patterns, two ASMAs will be considered: radii of 40 NM and 60 NM.

delay (Wu and Caves, 2004a). It also would help airlines to shape schedules, networks and aircraft utilization, and improve operational and financial efficiency (by designing the optimal turnaround time for ground operations).

Although the sample is rather heterogeneous (a range of 746 min), almost 67% of turnaround operations last less than 120 min. 34% of operations fall within the interval from 30 to 60 min. The mean, median and mode for the turnaround's length are 166 min, 78 min and 54 min respectively. For the purpose of our paper, aircraft turnaround activities are treated aggregately as a “single” process or “black box”. This approach provides us with a “macro” view of aircraft turnaround operations and simplifies the observation and modelling work needed to study interdependencies in the subsequent causal analysis (Wu, 2012). Also, this method informs us about the “standard” turnaround time needed for different clusters (types of fleets, airline operators, routes), as shown in Fig. 13. This allows revising the system's operational

behavior and the demand characteristics.

Before establishing a taxonomy for the system's level of saturation, a statistical analysis of the data allows us to observe some tendencies and learn some lessons about delay behavior and the system's response.

Fig. 14 (a) shows the demand profile against the declared capacity of the airport for the 22nd of July 2016, selected as the baseline day scenario following IATA's definition of *busy day* (IATA, 2016). Fig. 14 (b) depicts the accumulated hourly delay for arrival and departure operations. Meanwhile, Fig. 15 illustrates the evolution of (a) arrival and (b) departure average hourly delay over the day (with one standard deviation intervals), for the complete sample of operations. These figures show hours in Coordinated Universal Time (UTC). During summer (observation period), local time in Madrid is “UTC + 2 h”.

Fig. 14 shows that the arrival accumulated hourly delay presents two peak periods (11–13 and 19–21), with higher levels at the end of the day. Meanwhile, departure delay rises as traffic demand reaches

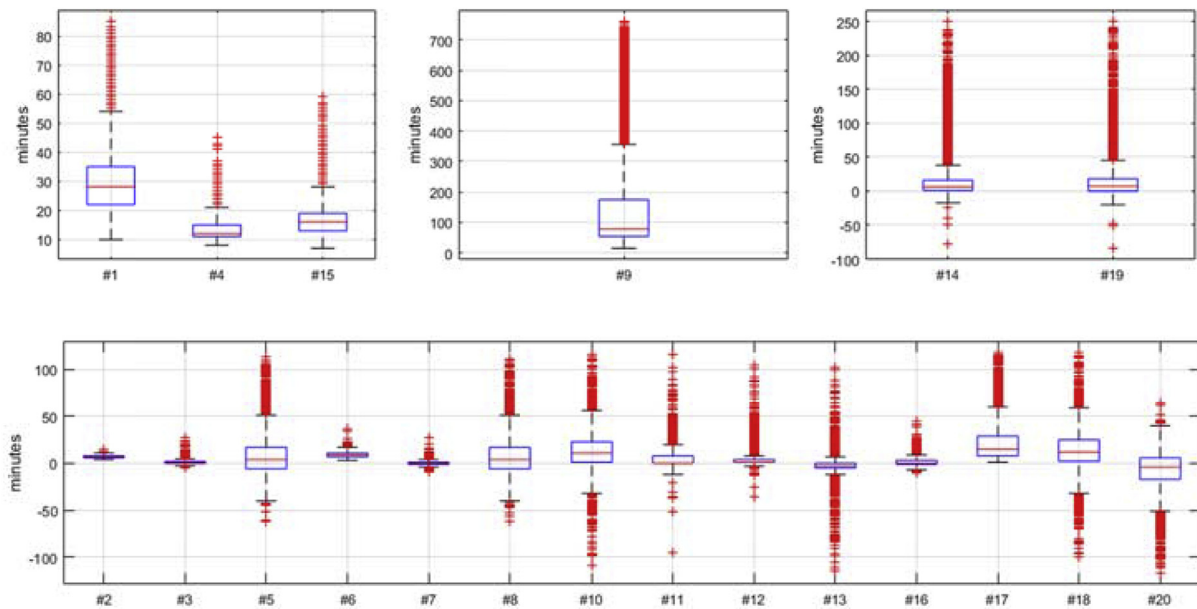


Fig. 10. Statistical characterization of the different items shown in Table 5.

capacity (near the hub operational windows). Moreover, Fig. 15 (a) illustrates that arrival average hourly delay increases and accumulates its impact over the day, due to the network effect. Meanwhile, departure average hourly delay does not follow this pattern, and presents two main peaks at early morning and late evening departures.

Fig. 16 (a) depicts the daily evolution of system and local delay (local delay = system delay - ATFCM delay). It shows the impact of airspace flow and capacity regulations on on-ground turnaround. Fig. 16 (b) illustrates that the effect of ATFCM delay is higher on the second part of the day.

Local delay does not completely follow arrival delay patterns, which implies that the airport-airspace system is at times somehow capable of absorbing a fraction of the arrival delay across the rotation stage. We analyze this aptitude by studying three mutually exclusive and complementary stages: taxi-in, turnaround and taxi-out (Fig. 17).

There is only a clear potential relationship between arrival delay and time recovery in the case of the turnaround stage (Fig. 17 (b)). Unimpeded taxi times can only be as short as the physics of the process allows, but can grow large in the event of a slow taxi operation (Simaiakis and Balakrishnan, 2015). Therefore, main opportunities to recover arrival delays arise at the turnaround stage. As discussed, turnaround duration can be “artificially” enlarged by the presence of buffers or ATFCM regulations. Fig. 18 (a) illustrates the system's ability to absorb delay and recover schedule adherence through the “pure” turnaround stage (without considering ATFCM delay). We focus on medium-short delays (between −60 min and +60 min delay in arrival), since for other thresholds the relationship is not so illustrative (causes for long delays are more related to irregular operations than to inefficiencies). Fig. 18 (b) depicts this ability for the baseline day

scenario.

We find a strong negative correlation ($R^2 = 0.733$) between arrival delay and turnaround delay, which is statistically significant at level 0.02 (p-value). The turnaround step is elastic enough to somehow adapt itself to arrival delay: when arrival delay increases in 1 min, turnaround delay decreases approximately in 0.8 min. The system responds to arrival delay through time buffers or by reducing operational times (improving efficiency), and hence on-ground turnaround is partially absorbing arrival delay. The elasticity of turnaround delay with respect to arrival delay depends on the type of operation (hour, airline, aircraft, route): Low Cost carriers (LCCs) at midday hours and Network carriers with high scheduled turnaround times present the higher potential for recovery. Negative values of arrival delay usually result in positive turnaround delay values due to slot adherence (LEMD is a coordinated airport).

Fig. 19 illustrates “pure” turnaround delay (without considering ATFCM delay) characteristics. Delay at the “pure” turnaround stage can adequately be represented by a Normal distribution ($\mu = 0.9$ min, $\sigma = 30.3$ min), being expressed with the following probability density function:

$$f_{NORMAL}(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot e^{-\left(\frac{x-\mu}{\sigma}\right)^2}$$

We can obtain the minimum recommended turnaround time for each aircraft type and configuration from its Airplane Characteristics for Airport Planning manual (see for example Boeing (2017)). By comparing this time with the scheduled turnaround time for each operation (and considering the average actual turnaround times for each airline and the ATFCM delay) we can estimate the implemented buffer

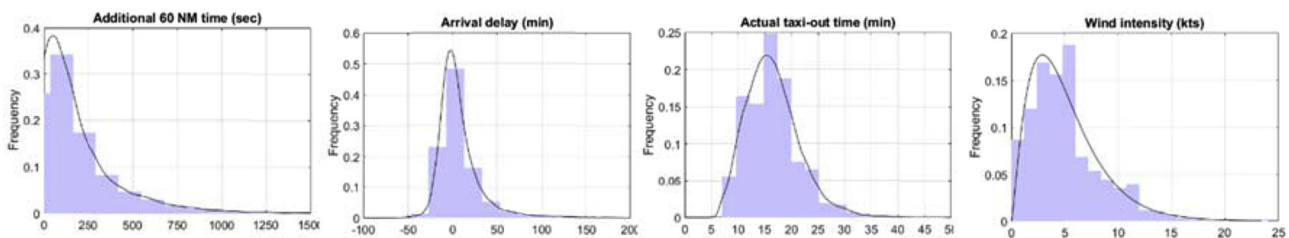


Fig. 11. Histogram and distribution fitting for (a) additional time at ASMA 60 NM (sec), (b) arrival delay (min), (c) actual taxi-out time (min), and (d) wind intensity (kts).

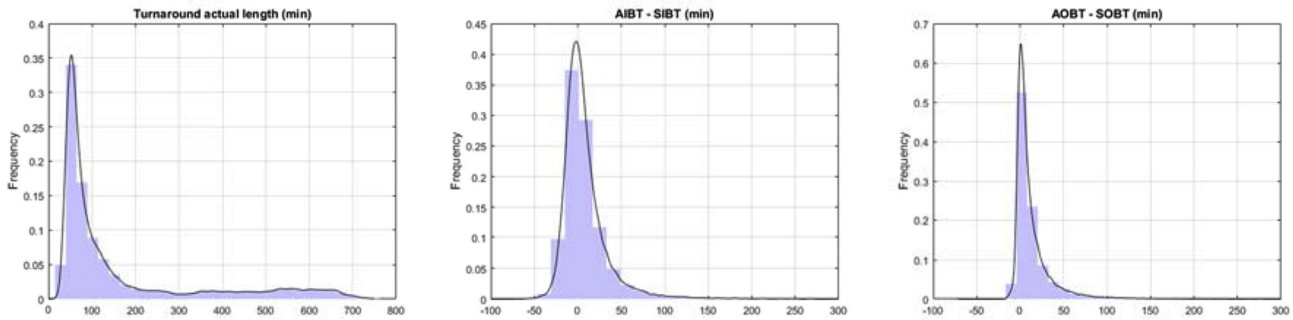


Fig. 12. Histogram and distribution fitting for turnaround: (a) actual duration (min), (b) starting time delay (AIBT - SIBT) (min), and (c) finishing time delay (AOBT - SOBT) (min).

time.

Data show that airlines often deploy long buffer times in parts of the day (when we can find higher scheduled turnaround times and lower local delays for the same type of operations) as a “fire break time”, in order to control delay propagation and its potential impact on schedule reliability. A proper scheduling of turnaround buffer time in aircraft rotations helps control the development of inherited delays in the network. It is a balance between tight turnaround schedules (which increase aircraft utilization by reducing flight connection time, but are too sensitive to disturbances in the network), and long turnaround schedules (which ensure off-block punctuality by adding “reserve times”, but reduce aircraft utilization and increase transfer times for passengers). Therefore, diverse airline business models present different challenges for airports. Data show that, except at midday, LCCs tend to schedule tighter turnaround times and hence find more difficulties for recovering arrival delay on the ground via buffer times (they need to be active in reducing processes’ times). Meanwhile, Network carriers stretch their scheduled turnaround times mainly during the hub operational windows; but outside these hours, they present enough “reserve time” for delay recovery.

Data do not demonstrate a clear positive relationship between delay (at the different stages) and the number of operations. Nevertheless, the correlation becomes stronger and statistically significant during congested hours (when the airport operates near its declared capacity). Fig. 20 shows that elasticity of “pure” turnaround delay with respect to arrival delay changes depending on the number of operations. The threshold of less than 20 operations/hour shows the higher potential for arrival delay recovery.

As can be seen in Fig. 21 (a), events of longer duration (arrival and turnaround processes) are the ones that contribute most to departure delay, and therefore to a potential operational saturation of the airport. These stages are the most likely to generate delays, but also those where more possibility of recovery exists. Delay at taxiing processes does not reach large levels in absolute terms compared to other stages, but it can be relatively significant in situations of congestion or at certain times: at the beginning and end of the day for taxi-in and midday for taxi-out

(Fig. 21 (b)).

Uncertainty of approach conditions makes traffic supply to runways a stochastic phenomenon. Hence, in order to ensure continuous traffic demand and maximize runway usage, a minimum level of queuing is required (Cook, 2007). Nevertheless, there is a trade-off between approach efficiency and runway throughput, because additional time in holding is harmful in terms of operational efficiency, fuel consumption and environmental impact. Fig. 22 (a) illustrates the average landing rate (aircraft/hour) as a function of the number of queuing aircraft at LEMD. It shows that the average landing rate stabilizes once there are approximately 21 queuing aircraft and stays at around 40–41 aircraft/hour. Therefore, any further approaches may lead to congestion and will not result in an improvement at the landing rate.

Symmetrically, if we look at the departure process, uncertainty of take-off conditions requires a minimum level of surface queuing to maximize runway usage. Pujet et al. (2000) evaluated surface congestion by considering the take-off rate of an airport as a function of the number of aircraft taxiing out. Fig. 22 (b) depicts this relationship for LEMD, where the departure rate stabilizes at around 37–38 aircraft/hour once there are approximately 36 departing aircraft on the ground (any further pushbacks may lead to congestion).

There is not a stable relationship between the number of aircraft queuing at arrival and the runway departure rate per hour, beyond the fact that the two of them increase during peak hours. This means that, at the observation period, arrivals and departures are not limiting each other, apart from the declared capacity and the operational procedures. Nevertheless, during high delay periods, we can find a cross-relationship that is statistically significant between arrival and departure processes, especially between the number of aircraft taxiing out and (a) the number of aircraft queuing for approach, (b) the number of flights with holding patterns and (c) the ASMA additional time.

Regarding airspace processes (E-TMA transit time, ASMA transit time and approach time), their lengths during non-congested hours result rather inelastic with respect to arrival delay. Meanwhile, during congested hours, data show a positive and statistically significant relationship between arrival delay and the number of aircraft queuing for

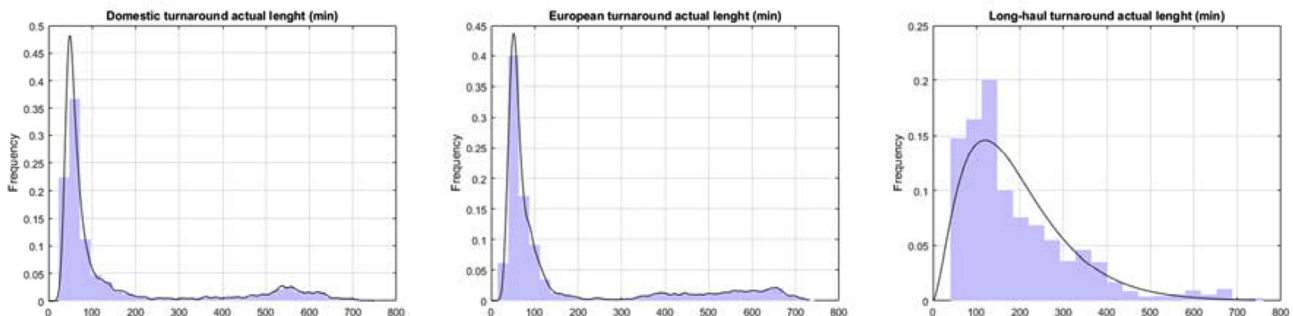


Fig. 13. Histogram and curve fitting for the turnaround time (min) at (a) Domestic, (b) European, and (c) Long Haul flights.

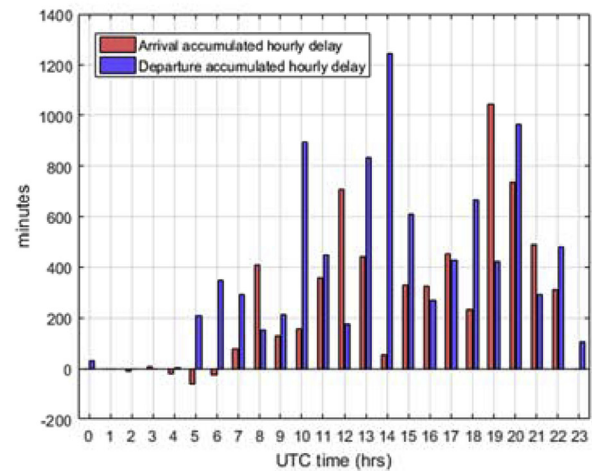
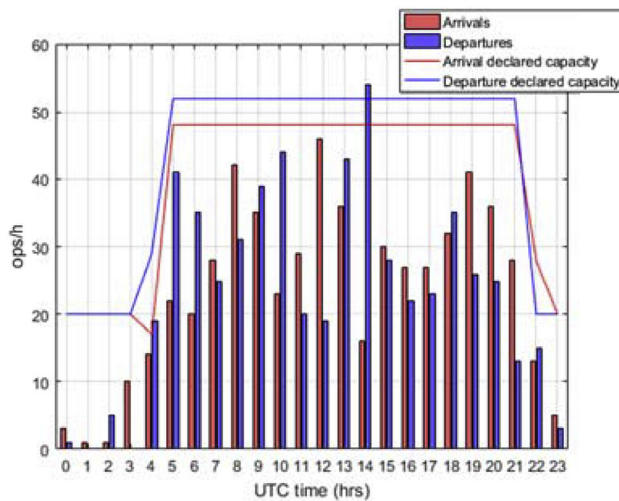


Fig. 14. (a) Traffic demand profile (22/07/2016) and (b) arrival/departure accumulated hourly delay profile (22/07/2016).

landing, and also between arrival delay and the number of flights with holding patterns. Therefore, the approach stage and its procedures prove to have substantial importance in reactionary delay and schedule adherence when the airport is operating near its declared capacity. Beyond the threshold of 20 arrivals/hour, arrival delay increases with the ASMA additional time, which in turn increases with the number of flights suffering holding patterns. The additional ASMA time and the number of flights with holding patterns almost stop increasing when the landing rate reaches 40 aircraft/hour. This phenomenon is related to the airport arrival capacity limit.

As expected, bad weather conditions (high wind intensity and low visibility) have a considerable and statistically significant impact on delays, especially on those related to airspace process: E-TMA transit time, ASMA transit time and approach time. This conclusion also arises when separation between aircraft (measured at the IAF or at the runway threshold) significantly increases (over one standard deviation) above average values.

Regarding the influence of airport layout, the highest actual taxiing times result from the combination of north runway configuration and the use of the T123 terminal area and its associated stands. Conversely, best opportunities for time-recovery at the taxiing stage (lowest taxiing delays) can be found with north configuration and the use of the T4/4S terminal area. Moreover, changes in runway configuration trigger an increment in actual taxiing times over the 30 min following the swift (delay is increased by 10% on average). Runway configuration is associated to the dominant wind direction and the use of certain aircraft

parking stands is related to the airline alliance that operates at each terminal area. Therefore, delay absorption strategies related to airport infrastructure use may not be always applicable. Nevertheless, data show that, within the same runway configuration, the use of different parallel runways can limit taxiing times (e.g. when T4T4S is assigned and the airport operates in north configuration, arrivals from runway 32R -and therefore approaches coming from the east airspace-suffer lower taxi-in delays).

Finally, Fig. 23 (a) shows the evolution of average hourly arrival delay throughout the day and the “future” average hourly system delay that aircraft arriving at each hour will present. Potential ability for delay recovery increases at the second half of the day. Nevertheless, it should be noted that delay recovery at the end of the day is mainly due to aircraft based at the airport. These operations present a long scheduled turnaround (no new legs programmed overnight) that enable them to recover arrival lateness and overcome the accumulated network effect, as can be seen in Fig. 23 (b).

3.3. Evaluation of the system's level of saturation

The operational efficiency of the airspace-airside system is critical in controlling delay propagation in an airline network, due to flight connections by aircraft routing (Bazargan, 2010). Fig. 24 (a) illustrates a graphical taxonomy for evaluating and characterizing the system's behavior in terms of time-efficiency and delay absorption “performance”. The horizontal axis represents the system delay, while the vertical axis

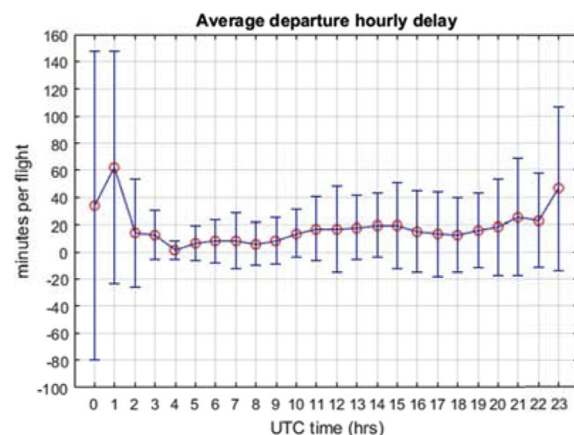
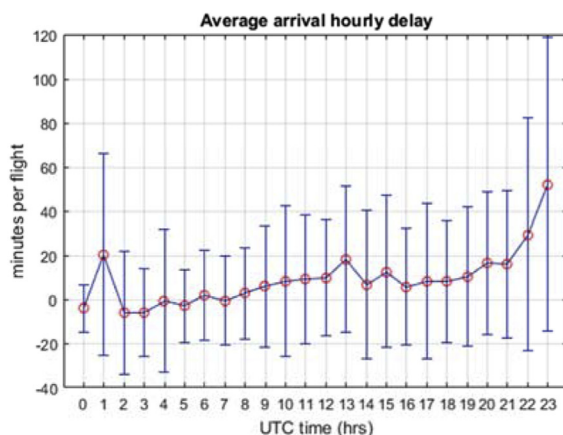


Fig. 15. Evolution of average hourly (a) arrival delay (min/op), and (b) departure delay (min/op) throughout the day (the error bars denote one standard deviation intervals).

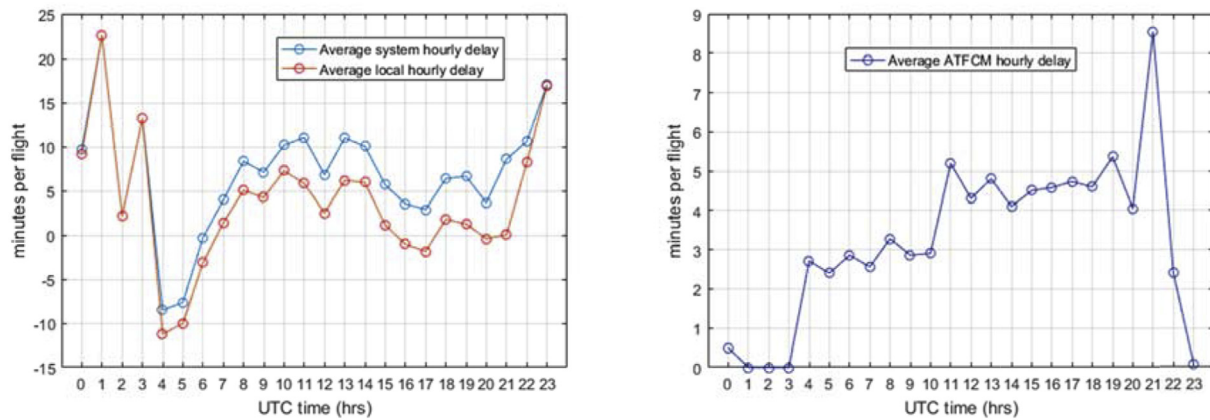


Fig. 16. Evolution of average system and local hourly delay (min/op) throughout the day, and (b) evolution of average ATFCM hourly delay (min/op) throughout the day.

represents the arrival delay. This phase map can be used to assess the “efficiency” of ground operations, i.e. the ability of the airport and the ground handling agents to respond to schedule perturbations. When operations are located in zone (1), the system is absorbing arrival delay and can mitigate it partially or even totally (departure delay develops at a lower rate than arrival delay). If operations fall in zones (2) or (3), it means that turnaround processes are further disturbed by operational disruptions or inadequacy of resources. Therefore, ground operations introduces new delay. It can be at a lower scale than arrival lateness (2) or even amplifying arrival delays (3). In zone (4), arrival delay is negative, but the system introduces schedule disturbances and generates delay, resulting in departure delayed operations (it could also be due to slot adjustments). Zone (5) implies that aircraft departs before scheduled time, which could not be possible (e.g. in coordinated airports with fixed allocated slots or due to passenger connections and crew changes). Fig. 24 (b) shows the system's behavior for the baseline day scenario (22/07/2016), with 36% of operations falling in zone (1), i.e. “delay absorption” phase.

As already mentioned, delay absorption “performance” can vary with the airline, handling agent, gate location, aircraft type, route origin and destination, local meteorological conditions, runway configuration, etc. Fig. 25 illustrates the graphical taxonomy for two different clusters during the observation period: (a) domestic flights operated by LCCs, which presents 24% of operations in zone (1) and (b) long haul flights operated by Network carriers, with 30% operations in zone (1). This result illustrates that tight turnarounds in domestic flights provide less opportunities for delay recovery.

This taxonomy helps us to obtain lessons about the potential for delay absorption, and the drivers that enhance it.

The operational behavior in terms of time saturation could also be analytically appraised through the algorithm shown in Fig. 26. It classifies the state of the system by relating arrival, system and departure delay. As previously stated, delay refers to the time difference between

the actual time and the schedule time of a timestamp or process. Therefore, delays can be positive or negative, meaning early departure or early arrival of a flight. Fig. 27 illustrates the evolution of the system's state (in terms of delay absorption) for the week of 18–24 July 2016, which includes the baseline day scenario. It is computed with the hourly average delay, but it could also be calculated for each operation. By analyzing this type of graphs for different clusters, we can obtain information about the system dynamics in terms of delay management performance, consecutive changes in delay phases and the probability of this to happen.

The development mechanism of departure delay and the system operational buffer can be extracted from Figs. 26 and 27. If arrival delay is shorter than the scheduled buffer time for the turnaround, then arrival delay will be partially or fully absorbed by the scheduled buffer time (*delay mitigation or absorption*), with the possibility of achieving schedule adherence. If arrival delay is larger than the buffer, the resulting departure delay may develop according to one of three scenarios. First, the system may be able to maintain efficient turnaround operations and the level of departure delay will be less than the one of inbound delay (*delay mitigation*). Second, if the system is able to perform without introducing additional delay, departure delay will reproduce the inbound schedule disturbance (*delay transmission*). In these previous situations, airlines may contain the scale of delay propagation in a network without tactically allocating more resources during operations. Third, ground operations may be further disturbed by the (a) late arrival of inbound aircraft, (b) late transfer of passengers, baggage and crew, (c) inefficient passenger/cargo terminal processes, or (d) disruptions in ground handling plans (*delay propagation or amplification*); hence, outbound departure flights suffer further delays, not only due to arrival lateness but also from additional disturbances during aircraft turnaround. The causal model developed later in this paper allows us to predict when the system will be facing this last situation and which conditions act as precursors for it. Once this situation is

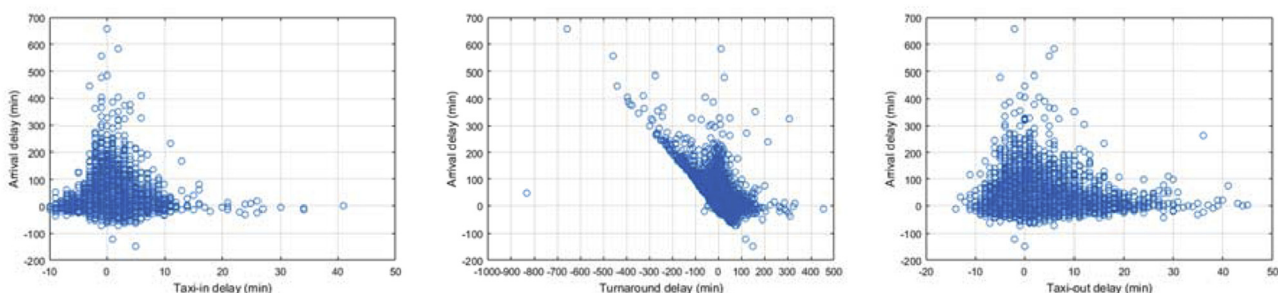


Fig. 17. Relationship between arrival delay and (a) taxi-in delay, (b) turnaround delay, (c) taxi-out delay (min).

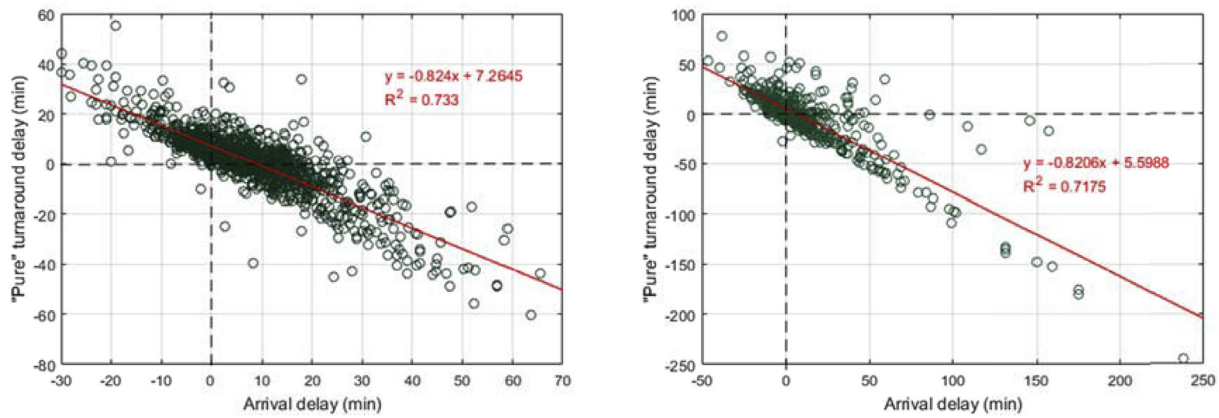


Fig. 18. Relationship between “pure” turnaround delay (without considering ATFCM delay) and arrival delay, (a) for the complete dataset (only short delays) and (b) for the baseline day scenario (22/07/2016).

predicted, delay could be partly compensated by means of enlarged buffers, improved ground operations efficiency or the allocation of more resources to speed up aircraft turnaround. These actions may stop or significantly reduce the risk of delay propagation in an airline network.

The delay behavior at each stage (arrival, taxi-in, turnaround, taxi-out) and the global level of saturation is also quantitatively appraised with the definition of several indexes and performance indicators (Table 6). These metrics evaluate delay evolution, compressibility of the processes, hourly distribution of delays throughout the day, contribution of each stage to the system's saturation and impact of different operations on delay propagation.

Fig. 28 and Fig. 29 show a selection of indexes for the baseline day scenario (22/07/2016). Fig. 28 (a) allows us to observe the daily evolution of departure delay and obtain some conclusions about its behavior (e.g. peaks are higher when approaching the hub operational windows, and they are usually followed by troughs, with the system recovering from congestion). Fig. 28 (b) illustrates the ability of the “pure” turnaround process (without considering ATFCM delay) to compensate arrival delay at certain times. Meanwhile, Fig. 29 (a) illustrates the hourly concentration of delays for each partial stage, when we assess delays by giving higher weight to heavy aircraft. It displays the impact of taxiing delays at certain times (especially midday). Fig. 29 (b) shows that average hourly departure delay does not reach its maximum values at most operated hours, but increases over the day (due to the accumulated effect of reactionary delay and network impact).

These taxonomies and indexes may allow policy makers, airport operators and air navigation service providers to evaluate the system's capacity to “receive and transmit” the expected aircraft flows with time efficiency. It also enables us to make time projections of future delays throughout the day.

3.4. Predicting departure delay and evaluating its precursors (a causal analysis)

Finally, the study is completed with a causal analysis through a Bayesian Network (BN) approach, which aims to understand the interdependencies between factors influencing performance and delay (drivers and predictors).

BNs are graphical probabilistic models used for reasoning under uncertainty (Pearl, 1993, 1986). This technique has proven to be an effective tool for risk assessment, resource allocation and decision analysis (Fenton and Neil, 2013). Moreover, BNs have unique strengths with respect to cross inference and visualization (Koller and Friedman, 2011) and have previously been used to tackle air transport issues (Bujor and Ranieri, 2016; Farr et al., 2013; Laskey et al., 2006; Morales-Napoles et al., 2006; Yorukoglu and Kayakutlu, 2011).

A BN is a directed acyclic graph (DAG), in which each node denotes a random variable, and each arc denotes a direct dependence between variables (nodes that are not connected symbolize variables that are conditionally independent of each other) (Kjærulff and Madsen, 2013). The DAG that results from the construction of a BN is quantified through a series of conditional probabilities based on data or

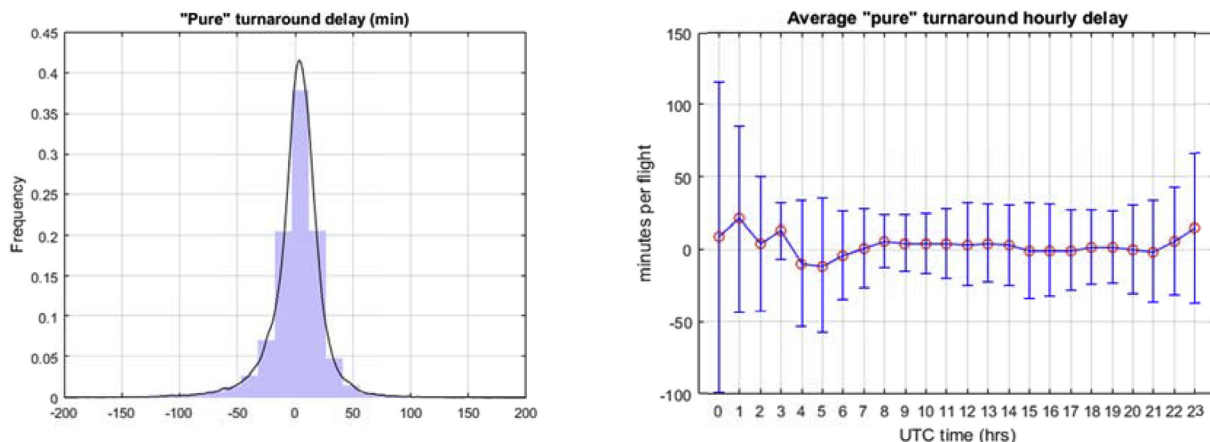


Fig. 19. “Pure” turnaround delay (without considering ATFCM delay): (a) histogram/curve fitting and (b) average hourly delay (min/op) throughout the day (the error bars denote one standard deviation intervals).

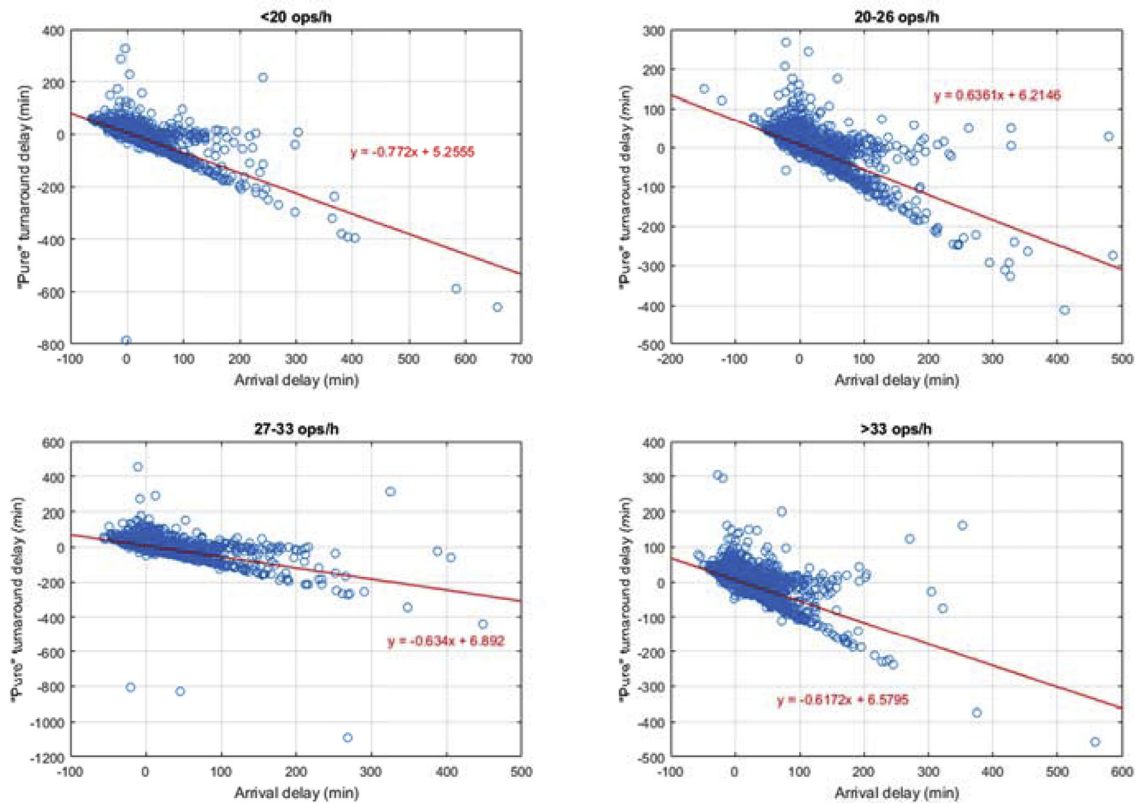


Fig. 20. Relationship between turnaround delay (without considering ATFCM delay) and arrival delay for different thresholds (number of operations).

information available on the system or problem. It defines a factorization of a joint probability distribution over the variables represented in the DAG (Darwiche, 2010; Koller and Friedman, 2011; Neapolitan, 2003). The factorization is represented by the directed links in the DAG (Nielsen and Jensen, 2007). That is, each node is associated with a probability function that takes (as input) a particular set of values for the node's parent variables, and gives (as output) the probability (or probability distribution, if applicable) of the variable represented by the node (Neapolitan, 2003). Therefore, the BN model structure (nodes and arcs) encodes conditional dependence relationships between the random variables. Each random variable is associated with a set of local probability distributions (parameters in the Conditional Probability Tables, CPT). Probability information in a BN is specified via these local distributions (Koller and Friedman, 2011). Consequently, a BN is a pair (G, P) , where G is DAG defined on a set of nodes x (the random variables), and $P = \{p(x_1 | \pi_1), \dots, p(x_n | \pi_n)\}$ is a set of n conditional probability densities (CPD), one for each variable. π_i is the set of

parents of node x_i in G . The set P defines the associated joint probability density of all nodes as (the chain rule for BN) (Kjærulff and Madsen, 2013; Pearl, 1993):

$$p(x) = p(x_1, \dots, x_n) = \prod_{i=1}^n p(x_i | \pi(x_i))$$

The graph G contains all the qualitative information about the relationships between the variables, no matter which probability values are assigned to them. Additionally, the probabilities in P contain quantitative information, i.e., they complement the qualitative properties revealed by the graphical structure (Darwiche, 2010; Koller and Friedman, 2011; Neapolitan, 2003).

Fig. 30 gives a basic example for a BN, where X_2 and X_3 are parent nodes of X_4 (child node for X_2 and X_3): the probability distribution of X_4 depends exclusively on the value of its parent variables (X_2 and X_3), i.e., X_4 is conditionally independent of X_1 given knowledge of X_2 and X_3 .

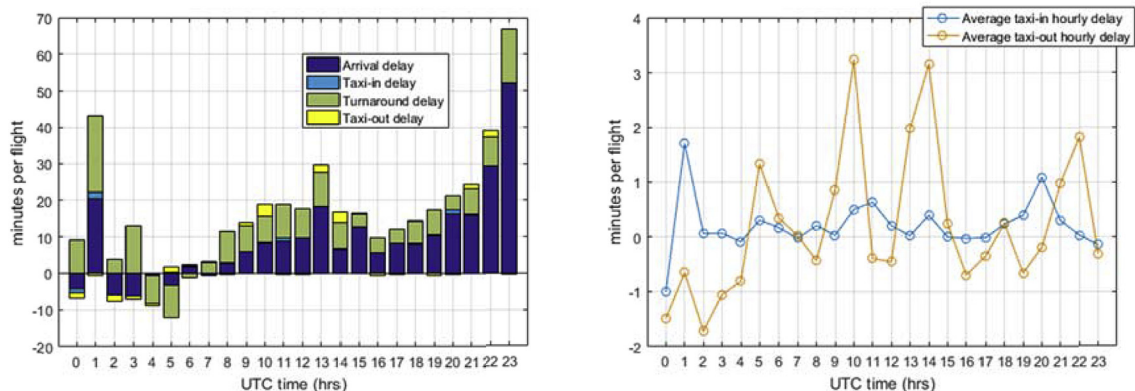


Fig. 21. (a) Contribution of each stage to departure hourly average delay (min/op), and (b) evolution of average taxiing hourly delay (min/op) throughout the day.

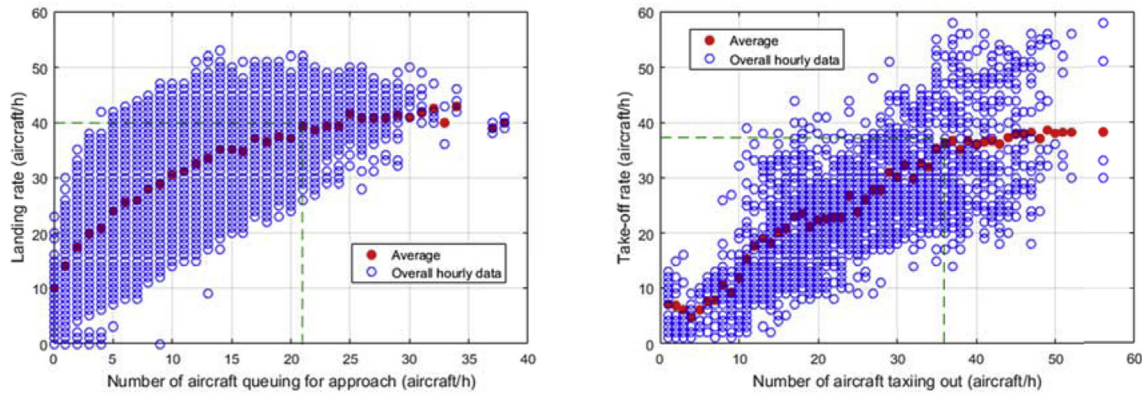


Fig. 22. (a) Average landing rate as a function of the number of queuing aircraft; and (b) average take-off rate as a function of the number of departing aircraft on the ground.

A BN can be constructed either manually, based on knowledge and experience acquired from previous studies and literature, or automatically from data (Kjærulff and Madsen, 2013). In this study, the selection of variables (Table 7) is constrained by the availability of data. We use the elements (timestamps, meteorological features, aircraft and airline data, flight details, operational characteristics and airport configuration) that have been analyzed and modeled throughout the study.

The first step in the BN construction process is to generate the correlation matrix for the variables involved, in order to assess the correlation among pairs (regression analysis). Nevertheless, correlation does not imply causation (Pearl, 2009). Although in a BN model it is not strictly necessary to have the directed links of the model following a causal interpretation, it just makes the model much more intuitive, eases the process of getting the dependence and independence relations right, and significantly facilitates the process of eliciting the conditional probabilities of the model (Kjærulff and Madsen, 2013). Therefore, proper modeling of causal relations (i.e. the directed links represent causal relations) is helpful for the model construction. For our purpose, causality is understood as follows: a variable X is said to be a direct cause of Y if setting the value of X by force, the value of Y may change and there is no other variable Z that is a direct cause of Y such that X is a direct cause of Z ; see Pearl's work for details (Pearl, 2009).

Subsequently, a data-driven process is employed to build the BN, applying a Bayesian Search (BS) algorithm (BayesFusion, 2017; Kjærulff and Madsen, 2013). This algorithm essentially follows a hill climbing procedure (guided by a scoring heuristic) with random restarts. It is proven to be highly effective in inter-causal propagation problems like the one we are facing (Cooper and Herskovits, 1992; Heckerman et al.,

1995). The BS algorithm uses the BDeu (Bayesian-Dirichlet equivalent uniform) function as the network scoring function in its search for the optimal graph. This scoring function is a common tool for selecting between different statistical models and it represents the goodness of fit of the model to observed data (Suzuki, 2017).

Discretization of variables is based on the statistical characterization previously developed. It is a crucial step to improve the model's accuracy, especially with “time” nodes (e. g., when a node represents delay at a process, discretization must ensure we neither lose information nor consider an excess of states).

The final step consists on validating the network structure with the judgment of key experts in the field (detailed in Table 1). The final architecture presented in Fig. 31 is hence determined by applying the BS algorithm (including variable discretization and validation) and then refining it with previous knowledge (inputs from the subject experts). Therefore, our model is built through a combination of a data-driven process and practical adjustments, in order to obtain a model reflecting reality. We develop a statistical significance test on pairs of nodes connected by an arc in the BN: associations between the nodes are statistically significant at level 0.02 (p-value).

Due to the conditional dependence relationships of the variables within the network, BNs offer the ability to either predict or diagnose (i.e. they can determine effects and causes). Therefore, two main scenarios reflect the utility of the model:

- Scenario 1 (forward/inter-causal scenario). The model predicts departure delay (output-child node) by setting the probability of having certain configuration, i.e. by setting one or more parent-

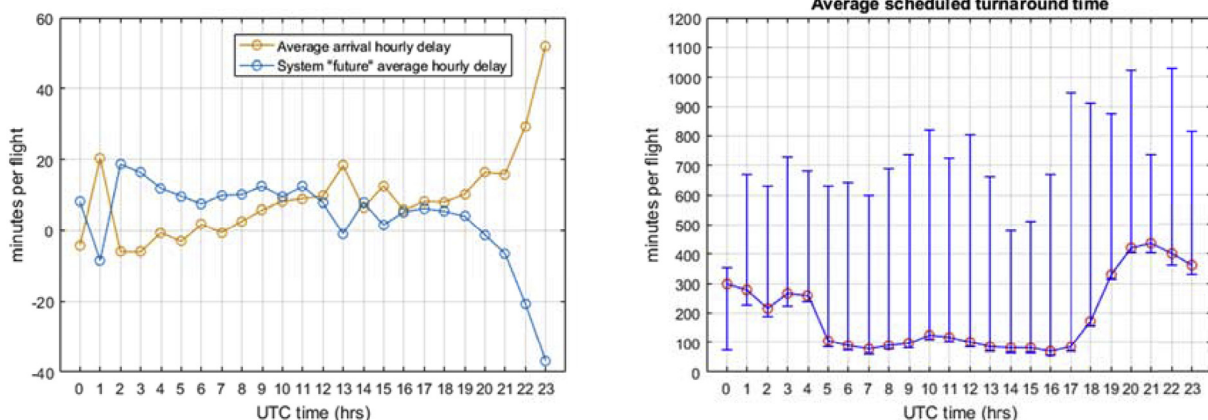


Fig. 23. (a) Average arrival hourly delay throughout the day and “future” system delay for flights arriving at each hour and (b) scheduled (min/op) turnaround length (the error bars denote the range).

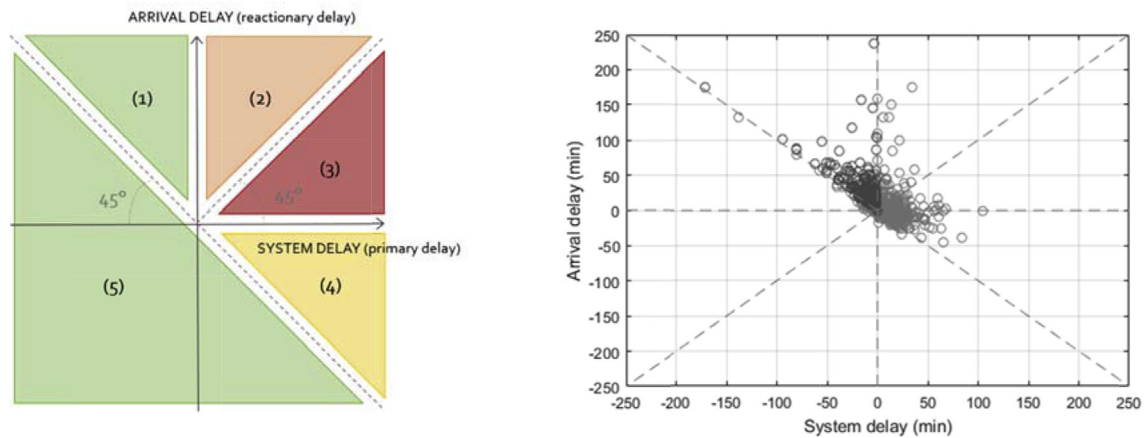


Fig. 24. Graphical taxonomy for the system behavior: (a) phase map and (b) application to the baseline day scenario (22/07/2016).

input nodes.

- Scenario 2 (backward inference). The model delivers a particular configuration in the parent nodes by setting the delay node to a target value. It provides understanding on which are the main contributors to delay (if delay is settled to a high positive value) or what configuration optimizes operations (if delay is settled to a negative value).

To illustrate the quantitative results obtained for scenario 1 and 2 at the BN model, two examples are presented below:

Scenario 1 (forward/inter-causal scenario). If we assign probabilities to particular events/states at several control nodes (e.g. route origin, terminal area, arrival runway, aircraft type and time frame for ALDT), the BN model infers the probability associated to each of the states of the node “departure delay” (Fig. 32). Hence, the node “departure delay” presents the discrete Probability Distribution shown in Fig. 32, when the probability of having an international flight at T4, arriving from runway 18L, operating a WB aircraft and landing in the morning is equal to one. Note that the node “departure delay” is discretized in the following states: less than −15 min, between −15 min and −3 min, between −3 min and 3 min, between 3 min and 15 min, and more than 15 min. For some “target” nodes (e.g. departure delay), the discretization is made not only attending to the actual data distribution but also to potential operational objectives (e.g. the ± 3 min threshold for punctuality set by SESAR’s performance metrics (Cook et al., 2013) and the 15 min threshold for defining delay that has historically been common to both Europe and the US (Cook et al., 2012a; Jetzki, 2009; Sherry et al., 2013)).

Scenario 2 (backward inference). If we establish the following

probabilities in the target node “departure delay” (Fig. 33): 25% for a delay below −15 min, 25% for a delay between −15 and −3 min and 50% for a delay between −3 and 3 min (i.e., 100% of having a departure delay below 3 min), we obtain the most probable states for the different parent nodes (e.g., route origin, terminal area, arrival runway, aircraft type and time frame for ALDT). In this case, the intervals with the highest probabilities are: 47% national (route origin), 78% T4 (terminal area), 83% 18R (arrival runway), 72% Wide Body (aircraft type) and 37% evening (13:00–20:00) (time frame for ALDT) (Fig. 33).

Hence, the model can be used as a decision making tool for resource allocation and for optimization of operational strategies regarding delay management. Nevertheless, it is not always possible to act on all the parameters (e.g. runway configuration is set by wind direction) or combine all the variables (e.g. terminal areas are assigned to specific airline alliances); i.e., some elements of the model are influence variables and not control variables.

A sub-sample of 90% of the observations is selected to train/build the model structure (i.e. establish the model’s ability to explain delay propagation). The remaining 10% of the data is set aside to test the accuracy of the predictions made by the model (i.e. test the model’s predictive capacity). The training and test sub-samples are randomly selected from the complete dataset.

The appraised scenarios provide promising results regarding the model’s ability to manage uncertainty (by explaining the system’s performance and predicting delay propagation). The test error ranges between 5% and 17%, with an average value of 10%. The highest test error appears in cases with extreme delay values, where there are less observations and the sample size is limited. This suggests that maybe over-sampling (i.e. deliberate selection of individuals of a rare type for

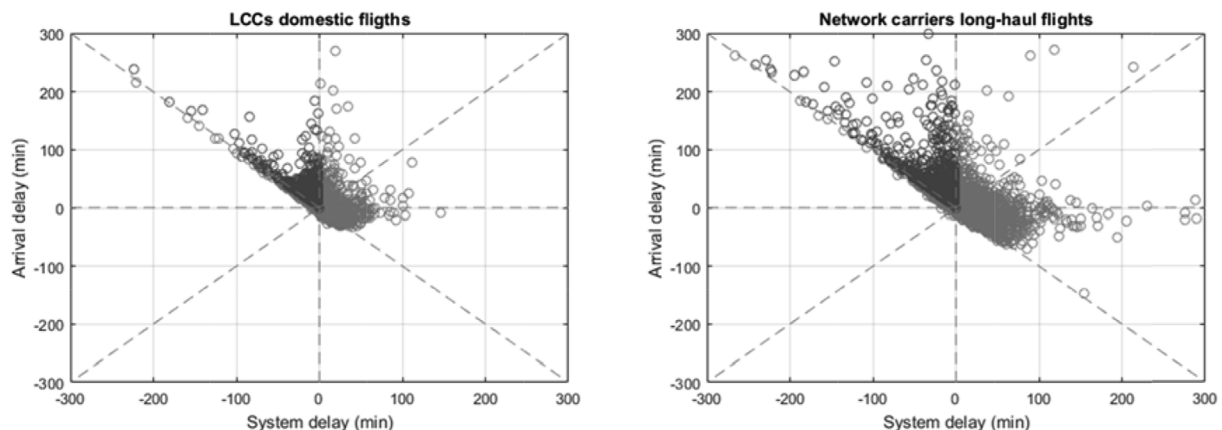


Fig. 25. Delay phase map for two types of operations: (a) Domestic flights by LCCs and (b) Long Haul flights by Network carriers.

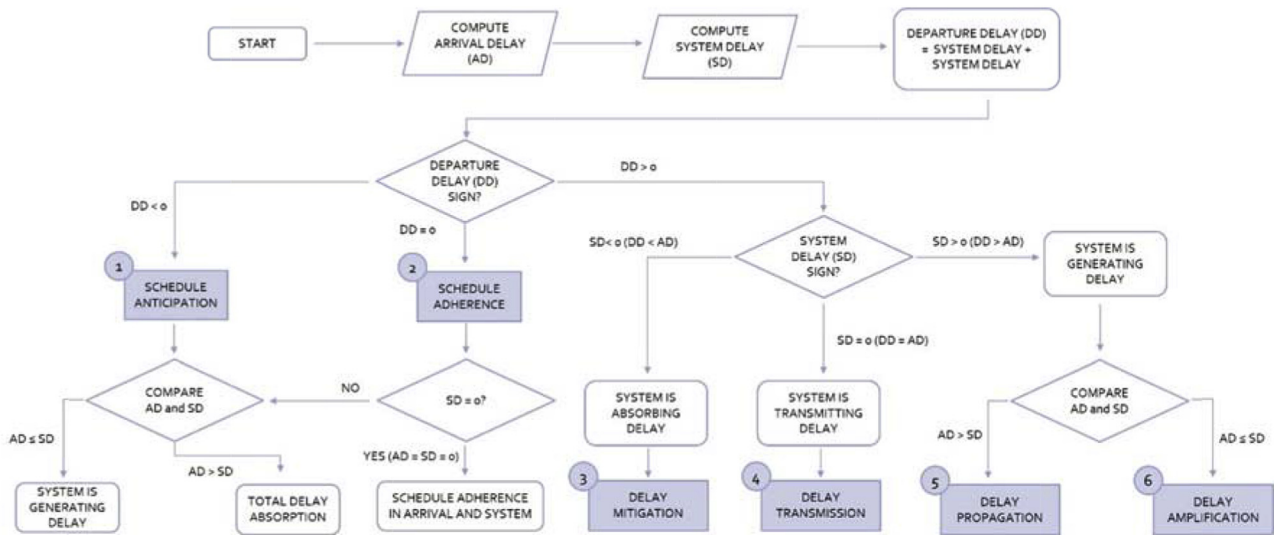


Fig. 26. Algorithm for evaluating the system's performance in terms of delay absorption.

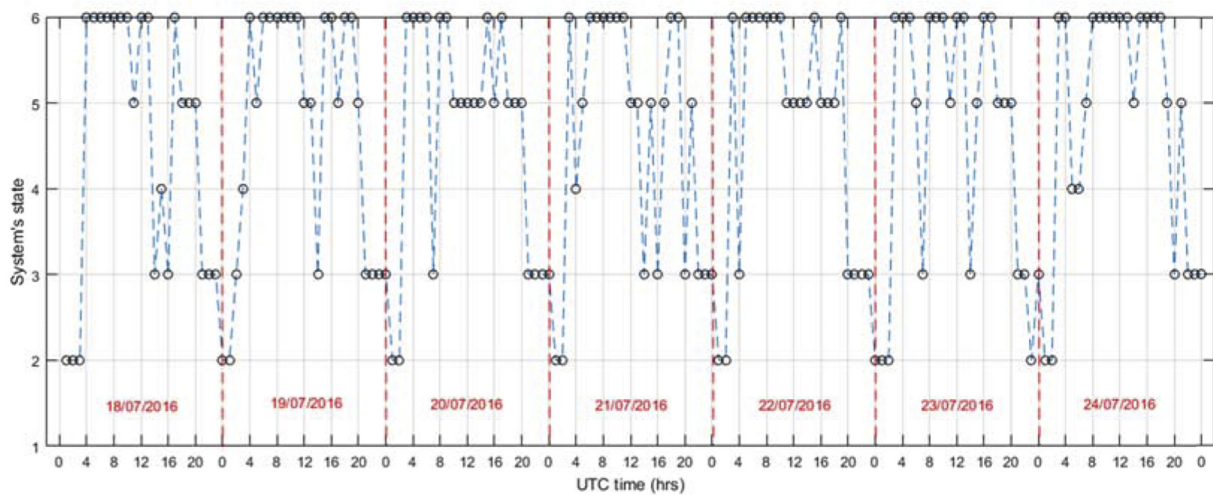


Fig. 27. Delay phase change through the week of 18–24 July 2016 (see system's state meaning in Fig. 26).

the training set), instead of random sampling, will be a good methodological improvement in future works. Moreover, the BN model is based on a probabilistic technique: the different nodes are discretized into different states/intervals, which present different associated probabilities. When comparing the actual data and the prediction provided by the BN model, there is an inherent “error” due to the size of the intervals. To adopt a conservative approach, we consider the difference between the actual observation and the furthest limit of the range. Fig. 34 shows a comparison of real data and BN predicted output for the variable “departure delay”. For the real data histogram, we use a Kernel density estimation approach to obtain a nonparametric representation of the probability density function of the variable (Scott, 2015).

The model allows us to obtain valuable conclusions regarding delay and time-saturation patterns (propagation dynamics, precursors and characteristics). Some of the lessons learned include information about:

- **Main delay triggers and their quantitative influence;** e.g. high amount of arrivals at early hours of the day, congestion at ASMA and E-TMA, bad weather conditions, tight scheduled duration of processes, LCCs operating domestic flights, changes in runway configuration, departure rates approaching runway capacity limit, and existence of external delay causes (traffic flow restrictions, aircraft technical

problems, crew schedule adherence requisites).

- **Ability of processes to compress themselves and achieve punctuality (potential for recovery);** e.g. overall turnaround delay decreases at a higher rate when the airport throughput is below the threshold of 20 operations/hour, taxiing processes reduce their delay at certain runway configurations, longer scheduled turnarounds at midday and late evening act as time-efficiency “protectors” for the system, unconstrained duration of airspace processes reaches a limit when the airport is operating near its declared capacity, and certain airlines are especially active in recovering arrival delay on the ground (reducing processes' time).

The quantitative influence of the different delay triggers can be assessed by defining thresholds in the values of those variables that cause a significant increase in the probability of having delay. For example, main departure delay precursors are: east wind (coming from the east and blowing toward the west) with an intensity over 15 kt, ASMA 60 NM additional time longer than 10 min, number of holdings above two, number of arriving aircraft (throughput for 40 NM and 60 NM over 40 aircraft), tight scheduled duration of processes (turnaround and taxi allocated times below the average for each operation type), departure rates reaching values above 80% of the runway capacity limit, and time frame (two main peaks at early morning and late

Table 6
Indexes and indicators for operational saturation analysis.

Index/Indicator	Equation	Definition and applicability
K_{ij} : Delay evolution indicator (applied to the total operation and to each of the partial stages)	$K_{ij} = \frac{D_{ij} - AD_{ij-20}}{ AD_{ij-20} } (\%)$	D_{ij} = Delay for flight i at stage j . AD_{ij-20} = Average delay over the 20 preceding flights to flight i (at stage j). It can be computed as a moving average, weighing the operations to assign a greater impact on the latest. If there are extreme delay values, the index provides better results with the geometric average. It measures delay evolution, considering the system's behavior throughout the last flights and allowing us to judge the efficiency of the actions implemented to solve the congestion problems.
DII_{ij} : Delay Impact Index	$DII_{ij} = \frac{D_{ij}}{PPL_{ij}} (\%)$	D_{ij} = Delay for flight i at process j . PPL_{ij} = Process planned length for flight i at process j . It measures the impact of delay over the expected duration of the process.
CI: Compressibility Index	$CI_{ij} = \frac{PAL_{ij}}{PPL_{ij}} (\%)$	PAL_{ij} = Process actual length for flight i at process j . PPL_{ij} = Process planned length for flight i at process j . It measures the process ability to be compressed.
θ_{ij} : Stage Contribution Index (to departure delay)	$\theta_{ij} = \frac{AHD_{ij}}{ AHD_{dj} } (\%)$	AHD_{ij} : Average hourly delay for stage i at hour j . AHD_{dj} : Average hourly departure delay at hour j . It measures the percentage of delay that each stage introduces at each hour, hence the main contributors can be detected.
α_1 and α_2 : Hourly Delay Performance Indicators, based on the aircraft type. A variation of the Herfindahl-Hirschman Index (Hirschman, 1964)	$\alpha_1 = \sum_k (\beta_k * AHD_{ij}^k) \left(\frac{\min}{op} \right)$ $\alpha_2 = \frac{\sum_k (\beta_k * AHD_{ij}^k)}{ ADD_i } (\%)$ (for each stage and each hour)	β_k = Weight of each aircraft wake-turbulence category. AHD_{ij}^k : Average hourly delay for stage i at hour j (for category k). NO_j = Number of operations at hour j . ADD_i : Average daily delay for stage i . Each type of aircraft, according to its wake-turbulence category (ICAO, 2016b), requires different operational procedures (with different levels of complexity) that may have an impact on the final delay. This index allows measuring the concentration of delays within each hour, referring to the type of operations served. The aircraft classification could be changed to reflect the influence of different variables: the engine category (turbofan/turboprop), the aircraft size (narrow/wide body) or the flight type (domestic/European/long haul).
η_1 and η_2 : Daily Delay Performance Indicators. A variation of the Lerner Index (Lerner, 1934)	$\gamma_j = \frac{NO_j}{\sum_i^{24} NO_j} (\%) \eta_1 = \sum_i^{24} \gamma_j * AHD_{ij} \left(\frac{\min}{op} \right)$ $\eta_2 = \frac{\sum_i^{24} \gamma_j * AHD_{ij}}{ ADD_i } (\%)$ (for each stage and each day)	NO_j = Number of operations at hour j . γ_j : Influence coefficient of each hour j . AHD_{ij} : Average hourly delay for stage i at hour j . ADD_i : Average daily delay for stage i . This indicator seeks to measure delay concentration throughout the day, assigning greater importance to the hours when the system has been more congested. This allows evaluating whether or not the actions taken during these periods of saturation have been successful in terms of reducing delay. A high value of γ with a low value of AHD_{ij} is a symptom of efficiency.

evening departure operations). When we have these operational conditions, the probability of having a departure delay over 15 min is higher than 60%. Also, low visibility conditions, LCCs operating a domestic or intra-European flight, south configuration (particularly departures from runway 14L) and aircraft belonging to a Medium wake

turbulence category are drivers for departure delay. Fig. 35 illustrates the impact of congestion at 60 NM (landing rate in the previous hour when the aircraft reaches this point) on departure delay for the aircraft's following leg. The node “congestion at 60 NM” (node 8 in Fig. 31) is discretized into five states (less than 10 aircraft, 10–20,

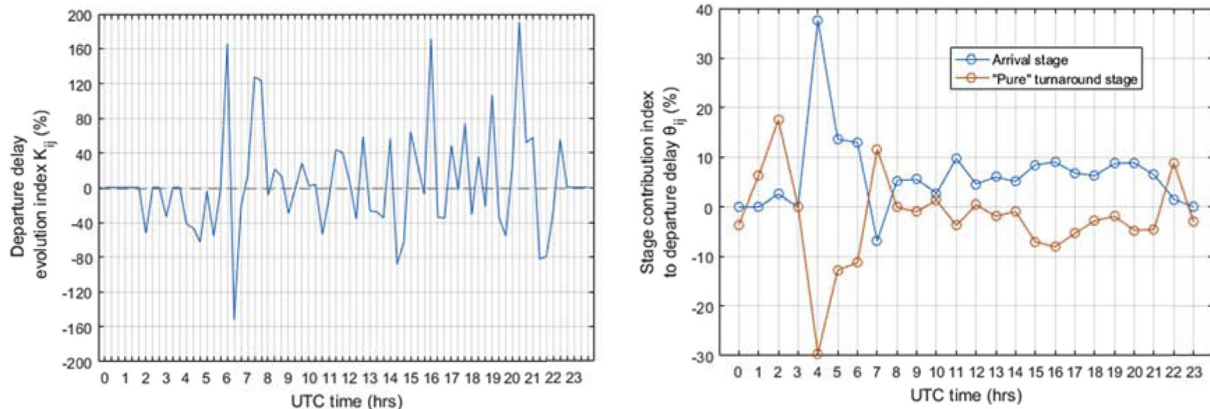


Fig. 28. (a) K_{ij} for departure delay and (b) θ_{ij} for arrival delay and “pure” turnaround delay (without considering ATFCM delay) (22/07/2016).

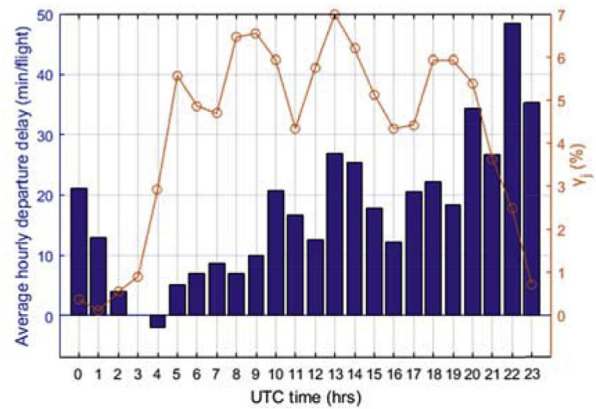
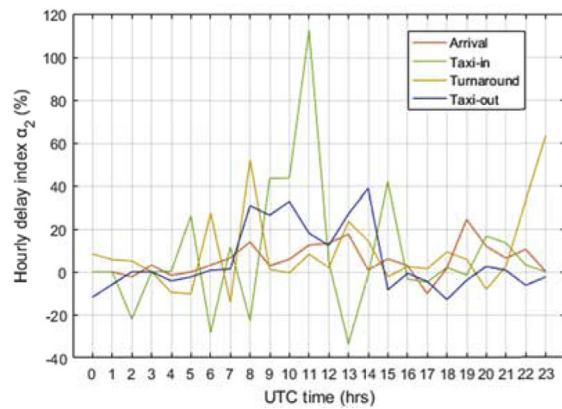


Fig. 29. (a) α_2 and (b) γ_1 against average hourly departure delay for the baseline day scenario (22/07/2016).

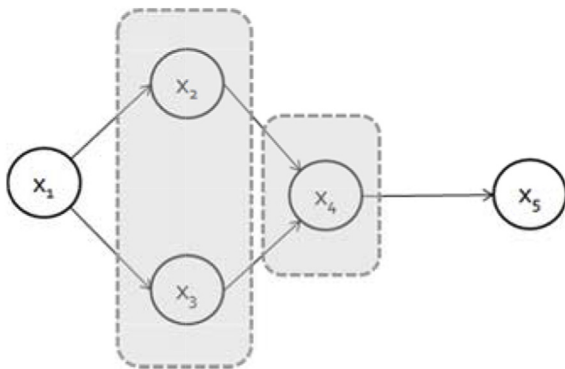


Fig. 30. Example of causal inference in a BN.

20–30, 30–40 and more than 40). With a forward analysis at the BN, when we assign a probability equal to one to these states, the model infers different probabilities for the diverse states of the node “departure delay” (node 43 in Fig. 31). For instance, when congestion at 60 NM is higher than 40 aircraft, there is a 32% probability of having a departure delay between 3 min and 15 min, and a 37% probability of obtaining more than 15 min of departure delay. Therefore, values over 40 aircraft for congestion at 60 NM are precursors for delayed departures.

Moreover, time for recovery can be appraised with the delay evolution index (k_{ij}) and the taxonomy of the system's state: after a severe saturation situation (continuous delays of more than 30 min), the system requires 120 min on average to recover time-efficiency and sustained punctuality. For each cluster (airline, aircraft, route, time of the day), saturation of scheduled time buffers occurs at a given arrival

Table 7

List of variables represented in the causal model (nodes at the BN).

Node number	Meaning	Node number	Meaning
1	Visibility	26	Arrival time - associated to ALDT (morning/afternoon/evening/night)
2	Wind direction	27	Departure time - associated to ATOT (morning/afternoon/evening/night)
3	Wind intensity	28	Arrival delay for the operation
4	Temperature	29	Scheduled taxi-in time
5	Amount of clouds	30	Actual taxi-in time (AIBT-ALDT)
6	Type of clouds	31	Taxi-in delay for the operation
7	Aircraft queuing (ASMA 60 NM)	32	Scheduled turnaround time (SOBT-SIBT)
8	Throughput (aircraft landed) in the previous hour, when aircraft reaches ASMA 60 NM	33	Actual turnaround time (AOBT-AIBT)
9	Aircraft queuing (ASMA 40 NM)	34	Actual “pure” turnaround time (without considering ATFCM delay)
10	Throughput (aircraft landed) in the previous hour, when aircraft reaches ASMA 40 NM	35	Turnaround delay for the operation
11	Additional ASMA time (60 NM)	36	“Pure” turnaround delay for the operation
12	Additional ASMA time (40 NM)	37	System delay for the operation
13	Number of holding patterns	38	Existence of ATFCM regulation for this flight
14	E-TMA transit time	39	ATFCM delay for the operation
15	Arrival configuration (north/south)	40	Scheduled taxi-out time
16	Arrival runway	41	Actual taxi-out time (ATOT-AOBT)
17	Departure configuration (north/south)	42	Taxi-out delay for the operation
18	Departure runway	43	Departure delay for the operation
19	Terminal area (T1/T2/T3/T4/T4S/cargo area/general aviation area) - associated to stand location (ramp)	44	Existence of delay according to the IATA coding system (IATA, 2015)
20	Type of airline operator (Low Cost/Network/Cargo/General Aviation)	45	Air Traffic Management related codes (IATA, 2015)
21	Aircraft size (narrow body/wide body)	46	Aircraft related delay codes (IATA, 2015)
22	Wake-turbulence category (H/M/L)	47	Airline related delay codes (IATA, 2015)
23	Handling agent	48	Airport related delay codes (IATA, 2015)
24	Route origin (domestic/European/long-haul)	49	Meteorology related delay codes (IATA, 2015)
25	Route destination (domestic/European/long-haul)	50	Other delay codes (IATA, 2015)
		51	System's state in terms of time-efficiency (saturation phase), see Fig. 26

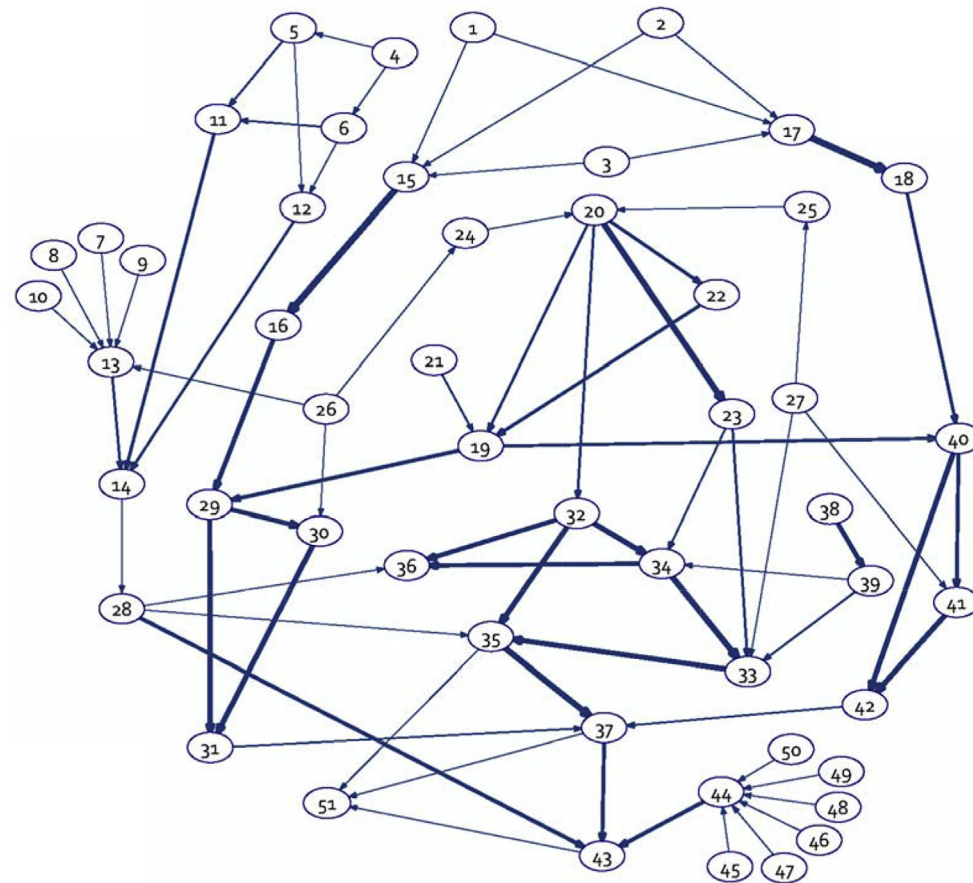


Fig. 31. BN model to understand the inter-dependencies between factors that influence delay performance and system saturation^(*). BN represented using GeNIe (BayesFusion, 2017).^(*) The thickness of an arc represents the strength of influence between two directly connected nodes. We use two measures of distance between distributions to validate results: Euclidean and Hellinger (Koiter, 2006).

delay intensity, resulting in departure delay increasing with arrival delay.

BNs have proven to be an excellent method to face the airport operational saturation problem for different reasons: (a) their construction allows combination between data and expert knowledge/judgment; (b) inference can be performed efficiently in models with a large amount of variables; (c) they develop a probabilistic approach to manage uncertainty and assess decision making, which is consistent with the treatment of stochastic processes; (d) they allow cross-inference (several control variables) for deriving conclusions under uncertainty, where multiple sources of information and complex interaction patterns are involved; and (e) they are “white boxes”, in the sense that the model components (variables, links, probability and utility parameters) are open to interpretation, which makes it possible to perform a whole

range of different analyses of the network (e.g., causal interactions, conflict analysis, (in)dependence analyses, sensitivity analysis, and value of information analysis).

4. Conclusions

This paper develops a functional analysis of the operations that represent the aircraft flow through the airport (airside)-airspace system. In this analysis, we use a dynamic spatial boundary associated with the E-TMA concept, so a linkage between inbound and outbound flights can be proposed and also airport and air traffic operations can be integrated. The aircraft flow is characterized by several temporal milestones related to the A-CDM method and structured by a hierarchical task analysis, providing a Business Process Model (BPM) for the rotation

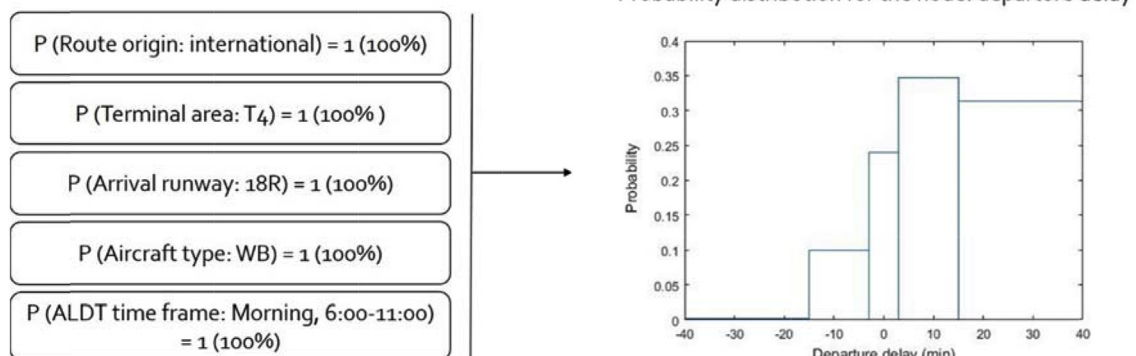


Fig. 32. Example of forward inference at the BN model.

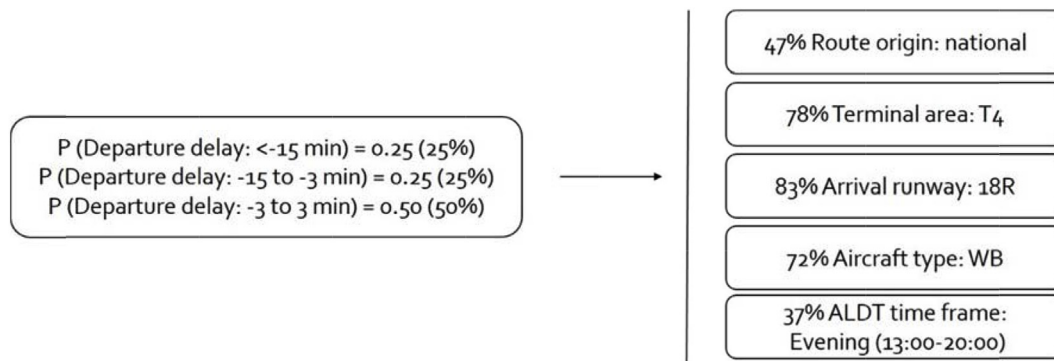


Fig. 33. Example of backward inference at the BN model.

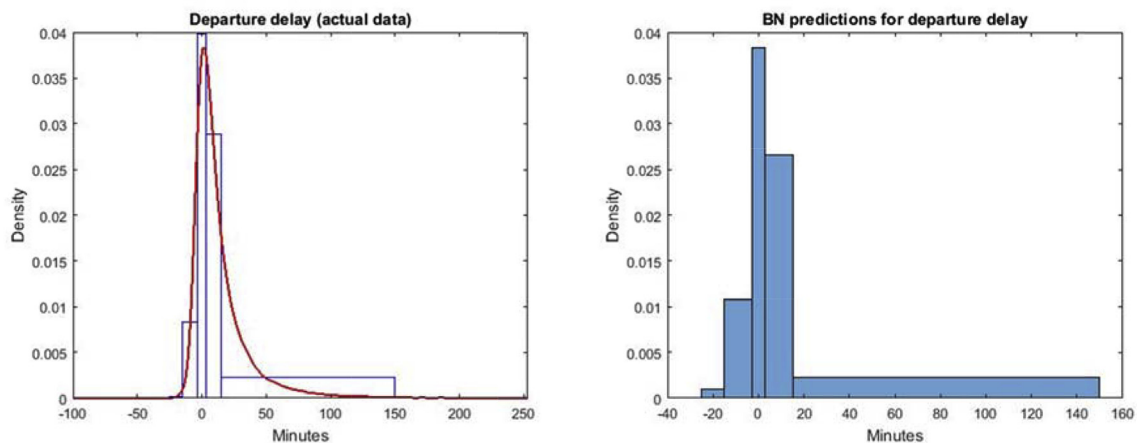


Fig. 34. Histograms for departure delay: (a) actual data; (b) BN predictions.

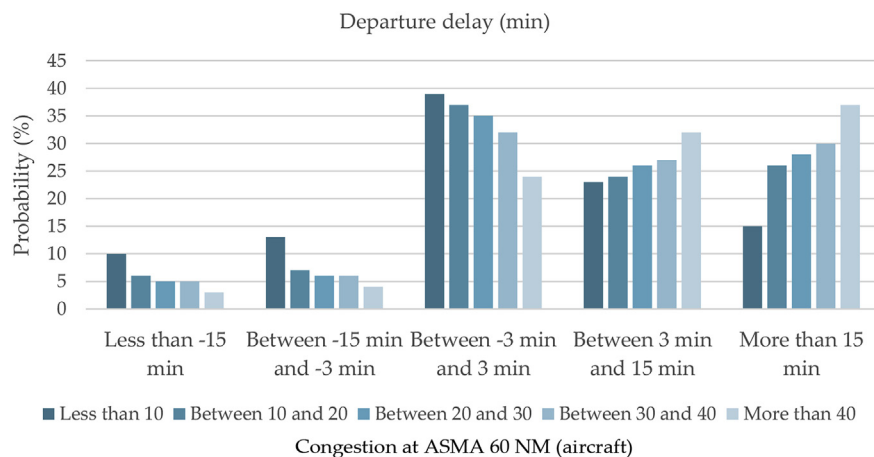


Fig. 35. Impact of ASMA 60 NM congestion on departure delay.

stage.

The application of the methodology to a case study of 34,000 turnarounds (registered at the peak traffic months of 2016) at Madrid Airport showed that arrival delay increases and accumulates its impact over the day, due to network effects. But local and departure delay do not follow this pattern, which implies that the airport-airspace system is somehow capable of absorbing a fraction of the arrival delay across the rotation stage. We analyze this aptitude by studying and characterizing the different processes that were previously identified with the BPM and the milestone approach. This evaluation of the system's level of saturation is completed by the proposal of a phase map and a taxonomy regarding delay absorption, and the definition of different indexes and

performance indicators.

Finally, the relationships between the factors that influence the aircraft flow are evaluated to create a probabilistic graphical model, using a Bayesian Network approach. This model predicts outbound delays given the probability of having different values at the causal control variables. Moreover, by setting a target to the output delay, the model provides the optimal configuration for the input nodes. Consequently, the model predicts and diagnoses delay propagation and system saturation, as a way to help inform policy makers and operators on appropriate measures for prevention and response.

Different lessons can be learned regarding delay propagation, time saturation and system recovery (where to act). Main potential drivers

for delay include time of the day, congestion at ASMA, weather conditions, amount of arrival delay, scheduled duration of processes, runway configuration, airline business model, handling agent, aircraft type, route origin/destination and existence of ATFCM regulations. Events with longer duration (arrival and turnaround processes) are the ones that contribute most to departure delay, but also those where more possibilities for recovery exist. In that sense, the model also informs us about how sensitive the different processes are to changes in arrival delay (arrival delay elasticity of processes).

The proposed methodology has several applications:

- Achieve a comprehensive understanding of operations at the E-TMA and relationships between variables.
- Investigate the impact of changes in tactical decisions and policies on the management and propagation of delays (“what-if” scenarios).
- The propagation model and the proposed taxonomy/indicators may be used to confirm that all agents collaborate in reducing delays, ensuring some target levels of efficiency.
- Using “forward” analysis it is possible to estimate the final departure delay and the system’s potential saturation, which can be used in the settlement of optimal turnaround and buffer times (sensitivity analysis).
- Using “backward” analysis it is possible to identify the main contributors (causes) to a final delay. This can be used to locate inefficiencies, diagnose irregular operations and define actions (allocate resources) that may correct the identified disruptions.

These applications address several challenges at the airport/airline industry: enhance system efficiency, improve robustness in the presence of operational uncertainties, ease resource allocation with multiple stakeholders and develop a decision-support tool that account for delay dynamics and behavior.

Future work will be focused on improving the accuracy and reliability of the model (more complete testing data and methodological improvements), and on comparing the results when the methodology is applied to other airports (generalize the case study). We also need to analyze potential response strategies/measures (reduce delays, mitigate inefficiencies and optimize operations), and apply the propagation model to other types of incidents (not just delays).

ANNEX I. List of symbols and abbreviations

Acronym	Meaning	Acronym	Meaning
A-CDM	Airport Collaborative Decision Making	ETOT	Estimated Take-Off Time
AD	Average Delay	(G, P)	Components of a BN: graph and probabilities
ADD	Average Daily Delay	H-M-L	Heavy-Medium-Light Turbulence Wake
AGHT	Actual Ground Handling Time	IAF	Initial Approach Fix
AHD	Average Hourly Delay	K	Delay evolution indicator
AIBT	Actual In-Block Time	KPI	Key Performance Indicator
ALDT	Actual Landing Time	LCC	Low Cost Carrier
AMAN	Arrival Management	LEMD	Adolfo Suárez Madrid Barajas Airport
ANSP	Air Navigation Service Provider	NB-WB	Narrow Body-Wide Body (aircraft size)
AOBT	Actual Off-Block Time	NM	Nautical Mile
ARDT	Aircraft Ready Time	NO	Number of Operations per hour

ASAT	Actual Start-Up Approval Time	PAL	Process Actual Length
ASMA	Arrival Sequencing and Metering Area	PPL	Process Planned Length
ASRT	Actual Start-Up Request Time	RVR	Runway Visual Range
ATC	Air Traffic Control	RWY	Runway
ATFCM	Air Traffic Flow and Capacity Management	SIBT	Scheduled In-Block Time
		SLDT	Scheduled Landing Time
ATM	Air Traffic Management	SOBT	Scheduled Off-Block Time
ATOT	Actual Take-Off time	STOT	Scheduled Take-Off Time
ATOT	Actual Take-Off Time	TDFT	Target Departure Fix time
ATV	Airport Transit View	TIAT	Target Initial Approach Fix Time
BDeu	Bayesian-Dirichlet equivalent uniform	TIBT	Target In-Block Time
BHS	Baggage Handling System	TLDT	Target Landing Time
BN	Bayesian Network	TMA	Terminal Maneuvering Area
BPM	Business Process Model	TOBT	Target Off-Block Time
BS	Bayesian Search Algorithm	TSAT	Target Start-Up Approval Time
CI	Compressibility Index	TTOT	Target Take-Off Time
CPD	Conditional Probability Density	TTOT	Target Take-Off Time
CPT	Conditional Probability Table	TWY	Taxiway
CTOT	Calculated Take-Off Time	UML	Unified Modelling Language
D	Delay	UTC	Coordinated Universal Time
DAG	Directed Acyclic Graph	α	Hourly Delay
DD	Departure Delay	β	Performance Indicator Weight of each aircraft wake-turbulence category
DII	Delay Impact Index	γ	Influence coefficient of each hour
EOBT	Estimated Off-Block Time	η	Daily Delay
E-TMA	Extended-Terminal Maneuvering Area	θ	Performance Indicator Stage Contribution Index

References

- Abdel-Aty, M., Lee, C., Bai, Y., Li, X., Michalak, M., 2007. Detecting periodic patterns of arrival delay. *J. Air Transp. Manag.* 13, 355–361. <http://dx.doi.org/10.1016/j.jairtraman.2007.06.002>.
- Abdelghany, K.F., Abdelghany, A.F., Ekollu, G., 2008. An integrated decision support tool for airlines schedule recovery during irregular operations. *Eur. J. Oper. Res.* 185, 825–848. <http://dx.doi.org/10.1016/j.ejor.2006.12.045>.
- Abdelghany, K.F., Shah, S.S., Raina, S., Abdelghany, A.F., 2004. A model for projecting flight delays during irregular operation conditions. *J. Air Transp. Manag.* 10, 385–394. <http://dx.doi.org/10.1016/j.jairtraman.2004.06.008>.
- AENA, 2017. Air Traffic Statistical Information [WWW Document]. URL: <http://www.aena.es/csee/Satellite?pagename=Estatisticas/Home> (accessed 11.17.17).
- AhmadBeygi, S., Cohn, A., Guan, Y., Belobaba, P., 2008. Analysis of the potential for delay propagation in passenger airline networks. *J. Air Transp. Manag.* 14, 221–236. <http://dx.doi.org/10.1016/j.jairtraman.2008.04.010>.
- Ashford, N.J., Stanton, H.P.M., Moore, C.A., Coutu, P., Beasley, J.R., 2013. *Airport Operations*, third ed. McGraw-Hill, New York.

- Bagieu, S., 2015. SESAR Solution Development, Unpacking SESAR Solutions, Extended AMAN. SESAR Joint Undertaking Solutions Workshop, Paris.
- Ball, M., Barnhart, C., Dresner, M., Hansen, M., Neels, K., Odoni, A., Peterson, E., Sherry, L., Trani, A., Zou, B., 2010. Total Delay Impact Study: a Comprehensive Assessment of the Costs and Impacts of Flight Delay in the United States. Washington D.C: NEXTOR Report Prepared for the Federal Aviation Administration (FAA).
- BayesFusion, 2017. GeNie Modeler.
- Bazargan, M., 2010. Airline Operations and Scheduling, second ed. Ashgate, Aldershot.
- Beatty, R., Hsu, R., Berry, L., Rome, J., 1999. Preliminary evaluation of flight delay propagation through an airline schedule. In: Air Track Control Quarterly, pp. 259–270. <http://dx.doi.org/10.2514/atcq.7.4.259>.
- Belkoura, S., Zanin, M., 2016. Phase changes in delay propagation networks. In: Proceedings of the 7th International Conference on Research in Air Transportation (ICRAT), Philadelphia, PA, USA.
- Belobaba, P., Odoni, A., Barnhart, C., 2015. The Global Airline Industry, second ed. <http://dx.doi.org/10.1002/9780470744734>. New York.
- Boeing, 2017. Airplane Characteristics for Airport Planning [WWW Document]. URL: http://www.boeing.com/commercial/airports/plan_manuals.page accessed 11.17.17.
- Bujor, A., Ranieri, A., 2016. Assessing air traffic management performance interdependencies through bayesian networks: preliminary applications and results. J. Traffic Transp. Eng. 4, 34–48. <http://dx.doi.org/10.17265/2328-2142/2016.01.005>.
- Buldyrev, S.V., Parshani, R., Paul, G., Stanley, H.E., Havlin, S., 2010. Catastrophic cascade of failures in interdependent networks. Nature 464, 1025–1028. <http://dx.doi.org/10.1038/nature08932>.
- Campanelli, B., Fleurquin, P., Arranz, A., Etxebarria, I., Ciruelos, C., Eguíluz, V.M., Ramasco, J.J., 2016. Comparing the modeling of delay propagation in the US and European air traffic networks. J. Air Transp. Manag. 56, 12–18. <http://dx.doi.org/10.1016/j.jairtraman.2016.03.017>.
- Campanelli, B., Fleurquin, P., Eguíluz, V.M., Ramasco, J.J., Arranz, A., Etxebarria, I., Ciruelos, C., 2014. Modeling reactionary delays in the european air transport. In: Proceedings of the 4th SESAR Innovation Days (SIDs), Madrid, Spain.
- Churchill, A., Lovell, D., Ball, M., 2007. Examining the temporal evolution of propagated delays at individual airports: case studies. In: Proceedings of the 7th USA/Europe Air Traffic Management Research and Development Seminar, Barcelona, Spain.
- Churchill, A.M., Vlachou, K., Lovell, D.J., 2008. Filtering and aggregation schemes for delay model calibration. In: Proceedings of the 3rd International Conference on Research in Air Transportation (ICRAT), Fairfax, Virginia, USA.
- Ciruelos, C., Arranz, A., Etxebarria, I., Peces, S., 2015. Modelling delay propagation trees for scheduled flights. In: Proceedings of the 11th USA/Europe Air Traffic Management Research and Development Seminar, Lisbon, Portugal.
- Cook, A., 2007. European Air Traffic Management: Principles, Practice and Research, first ed. Ashgate, Aldershot.
- Cook, A., Blom, H.A.P., Lillo, F., Mantegna, R.N., Micciché, S., Rivas, D., Vázquez, R., Zanin, M., 2015a. Applying complexity science to air traffic management. J. Air Transp. Manag. 42, 149–158. <http://dx.doi.org/10.1016/j.jairtraman.2014.09.011>.
- Cook, A., Tanner, G., 2014. European Airline Delay Cost Reference. University of Westminster commissioned by EUROCONTROL Performance Review Unit, Brussels.
- Cook, A., Tanner, G., Cristóbal, S., Zanin, M., 2012a. Passenger-oriented enhanced metrics. In: Proceedings of the 2nd SESAR Innovation Days (SIDs), Braunschweig, Germany.
- Cook, A., Tanner, G., Cristóbal, S., Zanin, M., 2015b. Delay propagation: new metrics, new insights. In: Proceedings of the 11th USA/Europe Air Traffic Management Research and Development Seminar, Lisbon, Portugal.
- Cook, A., Tanner, G., Lawes, A., 2012b. The hidden cost of airline unpunctuality. J. Transp. Econ. Policy 46, 157–173. <http://dx.doi.org/10.2307/24396360>.
- Cook, A., Tanner, G., Zannin, M., 2013. Towards superior air transport performance metrics – imperatives and methods. J. Aerosp. Oper. 2, 3–19. <http://dx.doi.org/10.3233/AOP-130032>.
- Cooper, G.F., Herskovits, E., 1992. A bayesian method for the induction of probabilistic networks from data. Mach. Learn. 9, 309–347. <http://dx.doi.org/10.1007/BF00994110>.
- Darwiche, A., 2010. Bayesian networks. Commun. ACM 53, 80–90. <http://dx.doi.org/10.1145/1859204.1859227>.
- Delgado, L., Prats, X., 2014. Operating cost based cruise speed reduction for ground delay programs: effect of scope length. Transp. Res. Part C Emerg. Technol. 48, 437–452. <http://dx.doi.org/10.1016/j.trc.2014.09.015>.
- Deshpande, V., Arkan, M., 2012. The impact of airline flight schedules on flight delays. Manuf. Serv. Oper. Manag. 14, 423–440. <http://dx.doi.org/10.1287/msom.1120.0379>.
- Deutschmann, A., 2012. Prediction of airport delays based on non-linear considerations of airport systems. In: Proceedings of the 28th International Congress of the Aeronautical Sciences (ICAS), Brisbane, Australia.
- Dobruszkes, F., Givoni, M., Vowles, T., 2017. Hello major airports, goodbye regional airports? Recent changes in European and US low-cost airline airport choice. J. Air Transp. Manag. 59, 50–62. <http://dx.doi.org/10.1016/j.jairtraman.2016.11.005>.
- ENAIRe, 2018. Aeronautical Information Publication (AIP). Aeronautical Information Service (AIS) [WWW Document]. URL: https://ais.enaire.es/aip/aipcd_en.aspx accessed 1.16.18.
- Engels, G., Förster, A., Heckel, R., Thöne, S., 2005. Process modeling using UML. In: Dumas, M., van der Aalst, W.M.P., ter Hofstede, A.H.M. (Eds.), Process Aware Information Systems: Bridging People and Software through Process Technology. Wiley & Sons, Chichester, pp. 85–118.
- EUROCONTROL, 2017. CODA (Central Office for Delay Analysis) Digest: All-causes Delay and Cancellations to Air Transport in Europe - 2016. European Organisation for the Safety of Air Navigation, Brussels.
- EUROCONTROL, 2016. A-CDM Impact Assessment. European Organisation for the Safety of Air Navigation, Brussels.
- EUROCONTROL, ACI, IATA, 2012. Airport CDM Implementation: the Manual. European Organisation for the Safety of Air Navigation, Brussels.
- Fageda, X., Flores-Fillol, R., 2016. How do airlines react to airport congestion? The role of networks. Reg. Sci. Urban Econ. 56, 73–81. <http://dx.doi.org/10.1016/j.regsciurbeco.2015.11.002>.
- Farr, A.C., Kleinschmidt, T., Johnson, S., Yarlagadda, P.K.D.V., Mengersen, K.L., 2013. Investigating effective wayfinding in airports: a Bayesian network approach. Transport 90–99. <http://dx.doi.org/10.3846/16484142.2014.898695>.
- Fenton, N., Neil, M., 2013. Risk Assessment and Decision Analysis with Bayesian Networks. CRC Press.
- Fleurquin, P., Campanelli, B., Eguíluz, V.M., Ramasco, J.J., 2014. Trees of reactionary delay: addressing the dynamical robustness of the us air transportation network. In: Proceedings of the 4th SESAR Innovation Days (SIDs), Spain, Madrid.
- Fricke, H., Schultz, M., 2009. Delay impacts onto turnaround performance. In: Proceedings of the 8th USA/Europe Air Traffic Management Research and Development Seminar, Napa, CA, USA.
- Goedeke, P., 2010. Networks in Aviation: Strategies and Structures, Networks in Aviation: Strategies and Structures. <http://dx.doi.org/10.1007/978-3-642-13764-8>.
- Gopalakrishnan, K., Balakrishnan, H., Jordan, R., 2016. Deconstructing delay dynamics: an air traffic delay example. In: Proceedings of the 7th International Conference on Research in Air Transportation (ICRAT), Philadelphia, PA, USA.
- Gulding, J., Knorr, D.A., Rose, M., Enaud, P., Hegendoerfer, H.P., 2013. Measuring air traffic management (ATM) delays related to airports: a comparison between the US and Europe. In: Modelling and Managing Airport Performance, pp. 67–93. <http://dx.doi.org/10.1002/9781118535844.ch3>.
- Heckerman, D., Geiger, D., Chickering, D.M., 1995. Learning bayesian networks: the combination of knowledge and statistical data. Mach. Learn. 20, 197–243. <http://dx.doi.org/10.1023/A:1022623210503>.
- Hirschman, A.O., 1964. The paternity of an index. Am. Econ. Rev. 54, 761.
- Hirst, M., 2008. The Air Transport System, first ed. Woodhead Publishing, Cambridge. <http://dx.doi.org/10.1533/9781845695224>.
- IATA, 2016. Airport Development Reference Manual 10th Edition. International Air Transport Association, Montreal.
- IATA, 2015. Airport Handling Manual 35th Edition. International Air Transport Association, Montreal.
- IBERIA, 2015. Flight Operations Manuals. IBERIA, Madrid.
- IBERIA, 2014. Ground Handling Operations Manuals. IBERIA, Madrid.
- ICAO, 2016a. Annex 14 to the Convention on International Civil Aviation, Volume 1: Aerodrome Design and Operations 7th Edition. International Aviation Civil Organization, Montreal.
- ICAO, 2016b. Document 8643: Aircraft Type Designators. International Civil Aviation Organization, Montreal.
- ICAO, 2009. Document 9883: Manual on Global Performance of the Air Navigation System, first ed. International Civil Aviation Organization, Montreal.
- ICAO, 2006a. Document 9157: Aerodrome Design Manual 3rd Edition. International Civil Aviation Organization, Montreal.
- ICAO, 2006b. Document 8168: Procedures for Air Navigation Services - Aircraft Operations 5th Edition. International Civil Aviation Organization, Montreal.
- ICAO, 2006c. Safety Oversight Manual. Part a 2nd Edition. International Civil Aviation Organization, Montreal.
- Janić, M., 1997. The flow management problem in air traffic control: a model of assigning priorities for landings at a congested airport. Transp. Plan. Technol. 20, 131–162. <http://dx.doi.org/10.1080/03081069708717585>.
- Jetzki, M., 2009. The Propagation of Air Transport Delays in Europe. Thesis in the Department of Airport and Air Transportation Research. RWTH Aachen University.
- Katsaros, A., Sánchez, A.C.S., Zerkowitz, S., Kerulo, B., Stark, N., Rekiel, J., Pascual, S.P., 2013. Limiting the impact of aircraft turnaround inefficiencies on airport and network operations: the TITAN project. J. Airt. Manag. 7, 289–317.
- Kazda, A., Caves, R.E., 2007. Airport Design and Operations, second ed. Elsevier, Amsterdam.
- Kirwan, B., Ainsworth, L.K., 1992. A Guide to Task Analysis: the Task Analysis Working Group, first ed. Taylor & Francis, London.
- Kjærulff, U.B., Madsen, A.L., 2013. Bayesian Networks and Influence Diagrams - a Guide to Construction and Analysis, second ed. Springer, New York. <http://dx.doi.org/10.1007/978-1-4614-5104-4>. Information Science and Statistics vol. 22.
- Koiter, J.R., 2006. Visualizing Inference in Bayesian Networks. Thesis in the Faculty of Electrical Engineering, Mathematics, and Computer Science, Department of Man-Machine Interaction. Delft University of Technology.
- Koller, D., Friedman, N., 2011. Probabilistic Graphical Models: Principles and Techniques, first ed. The MIT Press, Cambridge.
- Kosanke, L., Schultz, M., 2015. Key performance indicators for performance-based airport management from the perspective of airport operations. In: Proceedings of the 5th Air Transport and Operations Symposium (ATOS), Delft, Netherlands.
- Laskey, K.B., Xu, N., Chen, C.H., 2006. Propagation of delays in the national airspace system. In: Proceedings of the 22nd Conference on Uncertainty in Artificial Intelligence, Cambridge, Massachusetts, US.
- Lerner, A.P., 1934. The concept of monopoly and the measurement of monopoly power. Concept Monop. Meas. Monop. Power 1, 157–175.
- Liu, Y.-J., Cao, W.-D., Ma, S., 2008. Estimation of arrival flight delay and delay propagation in a busy hub-airport. In: Proceedings of the 4th International Conference on Natural Computation (ICNC). IEEE, Jinan, China. <http://dx.doi.org/10.1109/ICNC.2008.597>.
- Liu, Y.-J., Ma, S., 2008. Flight delay and delay propagation analysis based on bayesian network. In: Proceedings of 2008 International Symposium on Knowledge

- Acquisition and Modeling (KAM), pp. 318–322. <http://dx.doi.org/10.1109/KAM.2008.70>.
- Liu, Y., Wu, H., 2013. A remixed bayesian network based algorithm for flight delay estimating. *Int. J. Pure Appl. Math.* 85, 465–475. <http://dx.doi.org/10.12732/ijpam.v85i3.3>.
- Marintseva, K., Yun, G., Kachur, S., 2015. Resource allocation improvement in the tasks of airport ground handling operations. *Aviation* 19, 7–13. <http://dx.doi.org/10.3846/16487788.2015.1015291>.
- Morales-Napoles, O., Hanea, A., Kurowicka, D., Cooke, R.M., Roelen, A., 2006. Causal modelling in the Aviation Industry: an application for controlled flight into terrain. In: Guedes Soares, C., Zio, E. (Eds.), *Safety and Reliability for Managing Risk*. Taylor & Francis, London, pp. 1831–1838.
- Nash, J.M., 2008. Guidebook for Airport Irregular Operations (IROPS) Contingency Planning. Airport Cooperative Research Program (ACRP) Report 65. Washington DC: Transportation Research Board (TRB). <http://dx.doi.org/10.17226/14667>.
- Neapolitan, R.E., 2003. *Learning Bayesian Networks*, first ed. Prentice Hall, New Jersey.
- Neyshabouri, S., Sherry, L., 2014. Analysis of airport surface operations: a case-study of atlanta hartfield airport. In: *Proceedings of the Transportation Research Board 93rd Annual Meeting*, Washington DC, USA.
- Nielsen, T.D., Jensen, F.V., 2007. *Bayesian Network and Decision Graph*. Springer Science & Business Media <http://dx.doi.org/10.1007/978-0-387-68282-2>.
- Norin, A., Granberg, T.A., Yuan, D., Värbrand, P., 2012. Airport logistics - a case study of the turn-around process. *J. Air Transp. Manag.* 20, 31–34. <http://dx.doi.org/10.1016/j.jairtraman.2011.10.008>.
- Oreshko, B., Kunze, T., Gerbothe, T., Fricke, H., 2014. Turnaround prediction and controlling with microscopic process modelling GMAN proof of concept & possibilities to use microscopic process szenarios as control options. In: *Proceedings of the 6th International Conference on Research in Air Transportation (ICRAT)*, Istanbul, Turkey.
- Pearl, J., 2009. *Causality*, second ed. Cambridge University Press, New York.
- Pearl, J., 1993. Belief networks revisited. *Artif. Intell.* 59, 49–56. [http://dx.doi.org/10.1016/0004-3702\(93\)90169-C](http://dx.doi.org/10.1016/0004-3702(93)90169-C).
- Pearl, J., 1986. Fusion, propagation, and structuring in belief networks. *Artif. Intell.* 29, 241–288. [http://dx.doi.org/10.1016/0004-3702\(86\)90072-X](http://dx.doi.org/10.1016/0004-3702(86)90072-X).
- Plackett, R.L., 1983. Karl Pearson and the chi-squared test. *Int. Stat. Rev./Rev. Int. Stat.* 51, 59–72. <http://dx.doi.org/10.2307/1402731>.
- Prats, X., Hansen, M., 2011. Green Delay Programs: absorbing ATFM delay by flying at minimum fuel speed. In: *Proceedings of the 9th USA/Europe Air Traffic Management Research and Development Seminar*, Berlin, Germany.
- Pujet, N., Delcaire, B., Feron, E., 2000. Input-output modeling and control of the departure process of busy airports. *Air Traffic Control Q* 8, 1–32. <http://dx.doi.org/10.2514/atcq.8.1.1>.
- Pyrgiotis, N., Malone, K.M., Odoni, A., 2013. Modelling delay propagation within an airport network. *Transp. Res. Part C Emerg. Technol.* 27, 60–75. <http://dx.doi.org/10.1016/j.trc.2011.05.017>.
- Rakas, J., 2014. Defining and Measuring Aircraft Delay and Airport Capacity Thresholds. Airport Cooperative Research Program (ACRP) Report 104. Washington DC: Transportation Research Board (TRB). <http://dx.doi.org/10.17226/22428>.
- Rebollo, J.J., Balakrishnan, H., 2014. Characterization and prediction of air traffic delays. *Transp. Res. Part C Emerg. Technol.* 44, 231–241. <http://dx.doi.org/10.1016/j.trc.2014.04.007>.
- Redondi, R., Gudmundsson, S.V., 2016. Congestion spill effects of Heathrow and Frankfurt airports on connection traffic in European and Gulf hub airports. *Transp. Res. Part A Policy Pract.* 92, 287–297. <http://dx.doi.org/10.1016/j.tra.2016.06.011>.
- Sandrine, C., de L'epinay, I., Hustache, J., Jelinke, F., 2007. Environmental impact of air traffic flow management delays. In: *Proceedings of the 7th USA/Europe Air Traffic Management Research and Development Seminar*, Barcelona, Spain.
- Schaefer, L., Noam, T., 2003. Aircraft turnaround times for air traffic simulation analyses. In: *Proceedings of the Transportation Research Board 82nd Annual Meeting*, Washington DC, USA.
- Schmitt, D., Gollnick, V., 2016. *Air Transport System*, first ed. Springer-Verlag Wien, Vienna. <http://dx.doi.org/10.1007/978-3-7091-1880-1>.
- Scott, D.W., 2015. *Multivariate Density Estimation: Theory, Practice, and Visualization*, second ed. Wiley & Sons, New York Wiley Series in Probability and Statistics.
- SESAR, 2014a. In: *SESAR Concept of Operations Step 2 Edition 2014*. SESAR Joint Undertaking, Brussels (Ed. 01.01.00).
- SESAR, 2014b. *SESAR Performance Framework (Edition 2)*. Single European Sky ATM Research Consortium, Brussels.
- Sherry, L., Wang, D., Xu, N., Larson, M., Sherry, L., Xu, N., Larson, M., 2013. Statistical comparison of passenger trip delay and flight delay metrics. *J. Chem. Inf. Model.* 53, 1689–1699. <http://dx.doi.org/10.1017/CBO9781107415324.004>.
- Simaikis, I., Balakrishnan, H., 2015. A queueing model of the airport departure process. *Transp. Sci. trsc.* <http://dx.doi.org/10.1287/trsc.2015.0603>. 2015.0603.
- Simić Krstić, T., Babić, O., 2015. Airport traffic complexity and environment efficiency metrics for evaluation of ATM measures. *J. Air Transp. Manag.* 42, 260–271. <http://dx.doi.org/10.1016/j.jairtraman.2014.11.008>.
- Small, K.A., 1982. The scheduling of consumer activities: work trips. *Am. Econ. Rev.* <http://dx.doi.org/10.1126/science.151.3712.867-a>.
- Suau-Sanchez, P., Voltes-Dorta, A., Rodríguez-Déniz, H., 2016. Measuring the potential for self-connectivity in global air transport markets: implications for airports and airlines. *J. Transp. Geogr.* 57, 70–82. <http://dx.doi.org/10.1016/j.jtrangeo.2016.09.013>.
- Suzuki, J., 2017. A theoretical analysis of the BDeu scores in bayesian network structure learning. *Behaviormetrika* 44, 97–116. <http://dx.doi.org/10.1007/s41237-016-0006-4>.
- Trapote-Barreira, C., Deutschmann, A., Robusté, F., 2016. Managing airlines: the cost of complexity. *Transp. Res. Procedia* 18, 297–304. <http://dx.doi.org/10.1016/j.trpro.2016.12.039>.
- Tu, Y., Ball, M.O., Jank, W.S., 2008. Estimating flight departure delay distributions—a statistical approach with long-term trend and short-term pattern. *J. Am. Stat. Assoc.* 103, 112–125. <http://dx.doi.org/10.1198/016214507000000257>.
- Tukey, J.W., 1977. *Exploratory Data Analysis*, first ed. Pearson, London.
- United States Department of Transportation, 2017. *Airline On-time Statistics and Delay Causes*. Bureau of Transportation Statistics [WWW Document]. URL: <http://www.transtats.bts.gov/> accessed 11.17.17.
- Wang, P.T.R., Schaefer, L., Wojcik, L.A., 2003. Flight connections and their impacts on delay propagation. In: *Proceedings of the 22nd Digital Avionics Systems Conference (DASC)*, <http://dx.doi.org/10.1109/DASC.2003.1245858>. Indianapolis, IN, USA.
- Wilke, S., Majumdar, A., Ochieng, W.Y., 2014. Airport surface operations: a holistic framework for operations modeling and risk management. *Saf. Sci.* 63, 18–33. <http://dx.doi.org/10.1016/j.ssci.2013.10.015>.
- Wong, J.T., Tsai, S.C., 2012. A survival model for flight delay propagation. *J. Air Transp. Manag.* 23, 5–11. <http://dx.doi.org/10.1016/j.jairtraman.2012.01.016>.
- Wu, C.L., 2012. *Airline Operations and Delay Management: Insights from Airline Economics, Networks and Strategic Schedule Planning*, first ed. Ashgate Publishing, Surrey, England.
- Wu, C.L., 2008. Monitoring aircraft turnaround operations – framework development, application and implications for airline operations. *Transp. Plan. Technol.* 31, 215–228. <http://dx.doi.org/10.1080/03081060801948233>.
- Wu, C.L., 2005. Inherent delays and operational reliability of airline schedules. *J. Air Transp. Manag.* 11, 273–282. <http://dx.doi.org/10.1016/j.jairtraman.2005.01.005>.
- Wu, C.L., Caves, R.E., 2004a. Modelling and optimization of aircraft turnaround time at an airport. *Transp. Plan. Technol.* 27, 47–66. <http://dx.doi.org/10.1080/0308106042000184454>.
- Wu, C.L., Caves, R.E., 2004b. Modelling and simulation of aircraft turnaround operations at airports. *Transp. Plan. Technol.* 27, 25–46. <http://dx.doi.org/10.1080/0308106042000184445>.
- Xu, N., Donohue, G., Laskey, K.B., Chen, C., 2005. Estimation of delay propagation in the national aviation system using bayesian networks. In: *Proceedings of the 6th USA/Europe Air Traffic Management Research and Development Seminar*, Baltimore, MD, USA.
- Yorukoglu, M., Kayakutlu, G., 2011. Bayesian network scenarios to improve the aviation supply chain. In: *Proceedings of the World Congress on Engineering*, vol. 2 London, England.
- Zio, E., Sansavini, G., 2011. Modeling interdependent network systems for identifying cascade-safe operating margins. *IEEE Trans. Reliab.* 60, 94–101. <http://dx.doi.org/10.1109/TR.2010.2104211>.